# ROBSTRIDE 03

## 60N.M Quasi-Direct Drive Integrated Motor Module

## User Manual

**RobStride Dynamics**

https://www.robstride.com/

# RS03 instruction manual

## Precautions

1. Please use according to the working parameters specified in this article, otherwise it may cause serious damage to the product!
2. Do not switch the control mode when the joint is running. If you need to switch, send the command to stop the operation before switching.
3. Check whether the parts are in good condition before use. If the parts are missing or damaged, contact technical support in time.
4. Do not disassemble the motor at will, so as to avoid unrecoverable failure.
5. Ensure that there is no short circuit when the motor is connected, and the interface is correctly connected as required.

## Legal Statement

Before using this product, please read this manual carefully and operate the product according to the contents of this manual. If the user violates the contents of this manual to use this product, resulting in any property damage, personal injury accident, the company does not assume any responsibility. Because this product is composed of many parts, do not allow children to touch this product to avoid accidents. In order to prolong the service life of the product, do not use this product in high temperature and high pressure environment. This manual has been printed to the extent possible to include a description of the functions and instructions for use. However, due to the continuous improvement of product functions, design changes, etc., there may still be discrepancies with the products purchased by users.

The color and appearance of this manual may differ from the actual product. Please refer to the actual product. This manual is published by Beijing Lingfoot

Times Technology Co., LTD. (hereinafter referred to as Lingfoot), and Lingfoot may at any time make necessary improvements and changes to the inaccurate and up-to-date information in this manual, or make improvements to procedures and/or equipment. Such changes will be uploaded to the company's official website in electronic format. Details can be found in the download center (www.robstride.com). All images are for reference only. Please refer to actual objects.

# After-sales Policy

The after-sales service of this product is implemented in strict accordance with the Law of the People's Republic of China on the Protection of Consumer Rights and Interests and the Product Quality Law of the People's Republic of China. The service content is as follows:

1. Warranty period and contents

   a. Users who place orders on the online channel to purchase this product can enjoy the return service without reason within seven days from the day after signing. When returning goods, the user must present a valid proof of purchase and return the invoice. The user must ensure that the returned goods maintain the original quality and function, the appearance is intact, the trademarks and various logos of the goods themselves and accessories are complete, and if there are gifts, they should be returned together. If the goods are artificially damaged, artificially disassembled, missing packaging boxes, missing parts and accessories, they will not be returned. The logistics cost incurred during the return shall be borne by the user (see "After-sales Service Fee Standard"). If the user does not settle the logistics cost, it will be deducted from the refund amount according to the actual amount incurred. Refund the amount paid to the user within seven days from the date of receipt of the returned item. Refund method is the same as payment method. The

specific arrival date may be affected by factors such as banks and payment institutions.

b. The warranty period of this product is 1 year.

c. Within 7 days after the user signs for the next day, non-human damage performance failure occurs, through the Lingzhu after-sales service center test and confirmation, for the user to handle the return business, the user must present a valid purchase voucher, and return the invoice. Any freebies should be returned.

d. From 7 days to 15 days after the user signs for the next day, non-human damage performance failure occurs, through the Lingfoot after-sales service center test and confirmation, for the user to replace the whole set of goods. After the replacement, the three guarantee period of the goods themselves is recalculated.

e. From 15 days to 365 days after the user signed the next day, after the inspection and confirmation of the Lingfoot after-sales service center, it is a quality fault of the product itself, and can provide free maintenance services. The replacement of the faulty product is owned by Lingzu Company. The product is not faulty and will be returned as is. This product has been strictly tested after the factory, if there is a quality fault other than the product itself, we will have the right to refuse the user's return demand.

2. Non-warranty regulations The following circumstances are not covered by the warranty:

3. Exceed the warranty period specified in the warranty terms.

4. Failure to follow the instructions, resulting in product damage caused by wrong use.

5. Damage caused by improper operation, maintenance, installation, modification, testing and other improper use.

6. Non-quality failure caused by conventional mechanical loss, wear.

7. Damage caused by abnormal working conditions, including but not limited to falling, impact, liquid immersion, violent impact, etc.

8. Damage caused by natural disasters (such as floods, fires, lightning strikes, earthquakes, etc.) or incapacitated forces.

9. Damage caused by exceeding peak torque.

10. Damage caused by exceeding peak torque.

11. Failure or damage caused by other non-product design, technology, manufacturing, quality and other problems.

12. Use this product for commercial purposes.

In the case of the above situation, the user must pay the cost.

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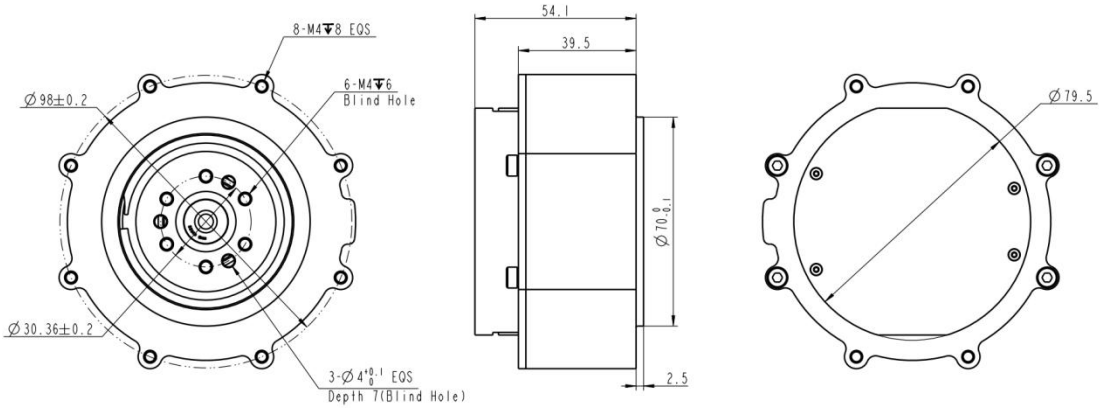
# 1. Motor specification

## 1.1. Outline and mounting dimensions



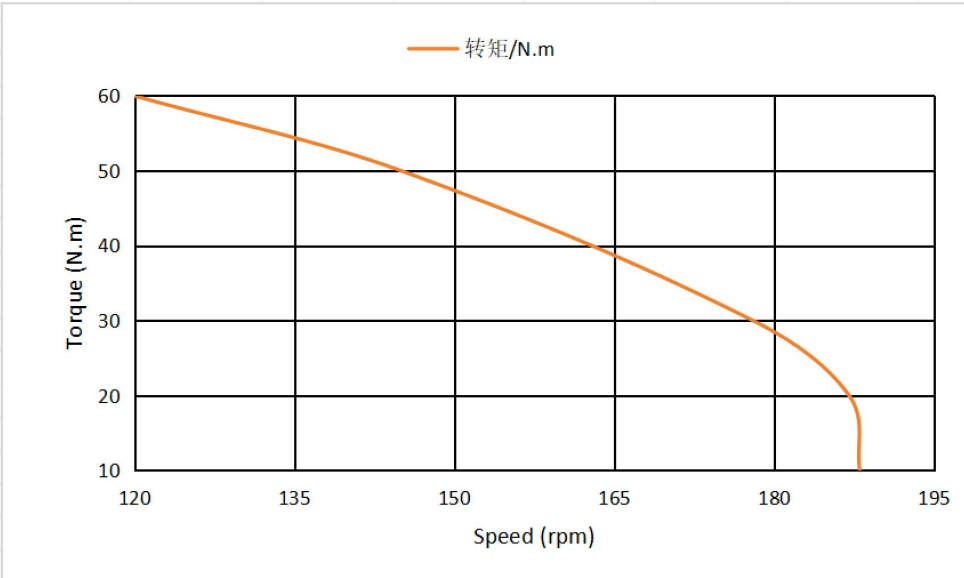

When fixing, the screw depth should not exceed the depth of the casing thread

## 1.2. Standard service condition

1. Rated voltage: 48 VDC
2. Operating voltage range: 24V-60 VDC
3. Rated load (CW) : 20 N.m(Heat sink dimensions: 215mm × 220mm)
4. Operation direction: CW/CCW from the direction of the exit shaft
5. Use posture: the direction of the exit axis is horizontal or vertical
6. Standard operating temperature: 25±5℃
7. Operating temperature range: -20 ~ 50℃
8. Standard operating humidity: 65%
9. Humidity range: 5 ~ 85%, no condensation
10. Storage temperature range: -30 ~ 70℃
11. Insulation Class: Class B

## 1.3. Electrical characteristics

1. No load speed: 200 rpm±10%
2. No load current: 2 Arms
3. Rated load: 20 N.m(Tested at 100 rpm)
5. Rated load phase current (peak) : 13Apk±10%
6. Peak load: 60 N.m
7. Maximum load phase current (peak) : 43Apk±10%
8. Insulation resistance/stator winding: DC 500VAC, 100M Ohms
9. High voltage/stator and housing: 600 VAC, 1s, 2mA
10. Motor back potential: 17Vrms/krpm±10%
11. Torque constant(Valid value): 2.36N.m/Arms
12. T-N curve




13. Maximum overload curve

Test Conditions:

Ambient Temperature: 25℃

Heater Material: Aluminum Alloy

Dimensions: 215mm × 220mm

Winding Limit Temperature: 145℃ (This is the constraint temperature; the actual temperature is 180℃)

Protection Logic:

Speed: 6nm tested at 100rpm

Test Data(Reduced heat sink size, lower rated torque)

| Loat | Operating time(s) |
|---|---|
| 60 | 13 |
| 50 | 34 |
| 40 | 67 |
| 30 | 393 |
| 20 | rated |

14. Thermal testing Test Conditions:

The winding temperature of a single phase is calculated in real time using the motor phase current and the motor thermistor temperature sensor. When the motor enters a stalled state, the three phases heat unevenly. The system calculates the temperature of each phase based on the current, and triggers protection when the calculated temperature reaches the thermal threshold.

PS: Under stall conditions, the heat generation of a single phase is 1.414 times that under rotating conditions. The stall overload time is as follows.

| Loat | Operating time(s) |
|---|---|
| 60 | 1 |
| 50 | 4 |
| 40 | 9 |
| 30 | 25 |
| 20 | 190 |
| 13 | rated |

## 1.4. Mechanical characteristic

1. Weight: 880g±20g
2. Number of poles: 42
3. Phase number: 3 phases
4. Drive mode: FOC
5. Deceleration ratio: 9:1

# 2. Driver Product Information

## 2.1. Driver product Specifications

| project | data |
|---|---|
| The rated working voltage | 48VDC |
| The maximum allowable voltage | 60VDC |
| Rated working phase current | 12Apk |
| Maximum allowable phase current | 43Apk |
| Standby power | ≤40mA |
| CAN bus bit rate | 1Mbps |
| Dimensions | Φ70mm |
| Working environment temperature | -20℃ to 50℃ |
| The maximum allowable temperature of the control board | 145℃ |
| encoder resolution | 14bit (absolute turn) |

## 2.2. Driver interface definition

## 2.3. Driver interface recommended brand and model

| board end model | brand manufacturer | line end model | brand manufacturer |
|---|---|---|---|
| XT30APW-M | AMASS (Ams) | XT30UW-F | AMASS (AMS) |
| GH1.25-2PWT | any | GH1.25-T | any |

## 2.4. Driver function pin and device description

1. Power supply and CAN communication

| Pin | description |
|---|---|
| 1 | The positive electrode of the power supply (+) |
| 2 | Negative electrode of the power supply (-) |

| Pin | description |
|---|---|
| 3 | CAN CAN_L |
| 4 | CAN the high side of the communication CAN_H |

1. Download port

| Pin | description |
|---|---|
| 1 | SWDIO (data) |
| 2 | SWCLK (clock) |
| 3 | 3V3 (positive 3.3V) |
| 4 | GND |

1. Indicator light

Driver function pin and device description

# 3. Upper computer instructions

Please go to www.robstride.com website download center

## 3.1. Hardware disposition

The articulated motor uses the CAN communication mode and has two communication cables. It is connected to the debugger through the can to USB tool. The debugger needs to be installed with the ch340 driver in advance and works in AT mode by default.

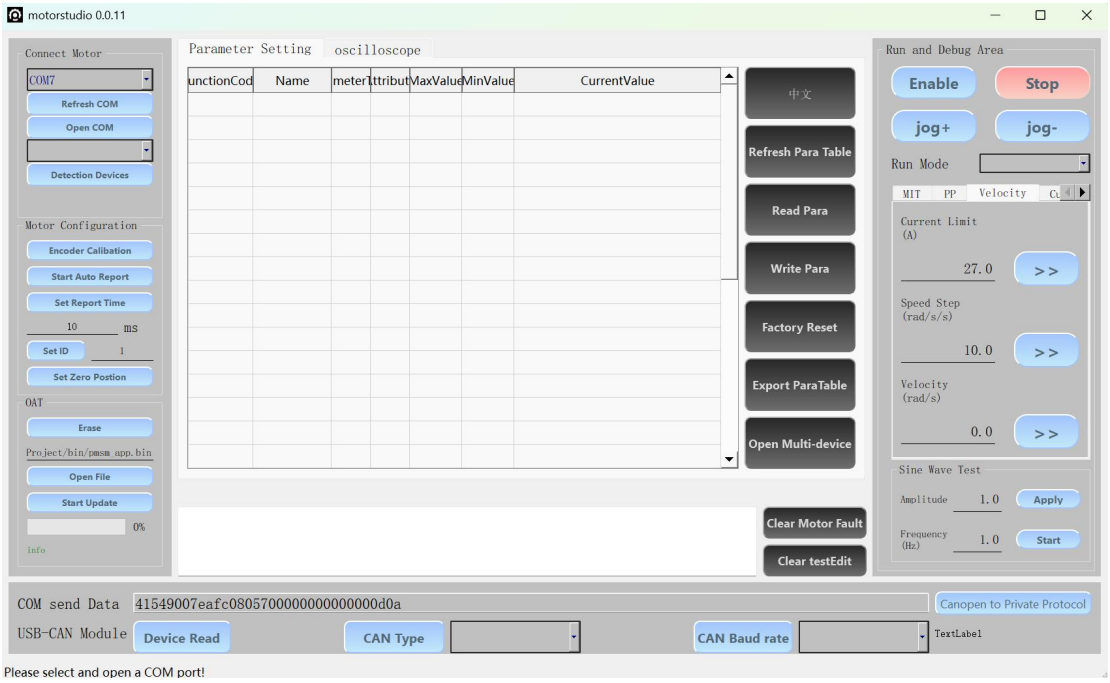
It should be noted that we are based on the specific can to USB tool development of the debugger, so we need to use our recommended serial port tool to debug the debugger, if you want to transplant to other debugger platform can refer to the third chapter of the instructions for development.

The CAN to USB tool is recommended to use the official USB-CAN module of Lingzu Times. The frame header of the corresponding serial port protocol is 41 54, and the frame tail is 0D 0A.

When using the CAN-to-USB module, pay attention to the settings of the DIP switches on the module: When DIP switch 1 is in the ON position, the module

enters Boot mode and cannot establish a connection with the host computer. When DIP switch 2 is in the ON position, a 120Ω terminal resistor is connected to the module port, allowing normal communication with the host computer.

## 3.2. Upper computer interface and description




Mainly includes:

A. Motor Connection Module

- Refreshing the Serial Port
- Opening the Serial Port
- Testing the Device

B. Motor Configuration Module

- Starting the Upgrade
- Opening a File
- Starting the Upgrade
- Modifying the Motor CAN ID
- Setting the Motor's Mechanical Zero Position

C. Motor Upgrade Module

- Magnetic Encoder Calibration
- Motor Active Reporting Switch
- Setting the Motor Active Reporting Time
- Modifying the Motor CAN ID
- Setting the Motor's Mechanical Zero Position

D. Motor Main Interface

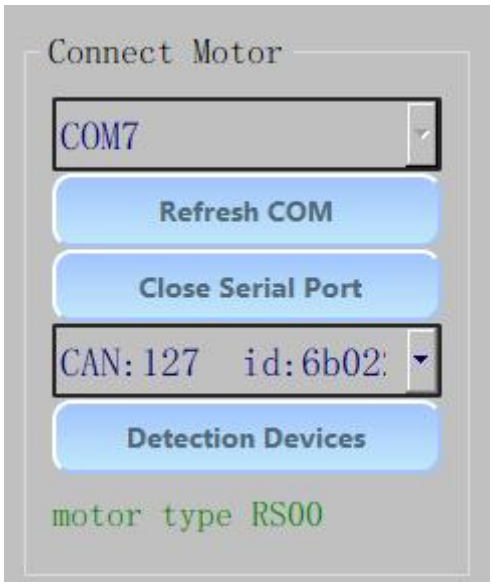
- Parameter Settings
- Motor Oscilloscope

E. Run and Debug Area

- Parameter Debugging Buttons
- Motor Mode Configuration and Parameter Modification
- Sine Signal Testing

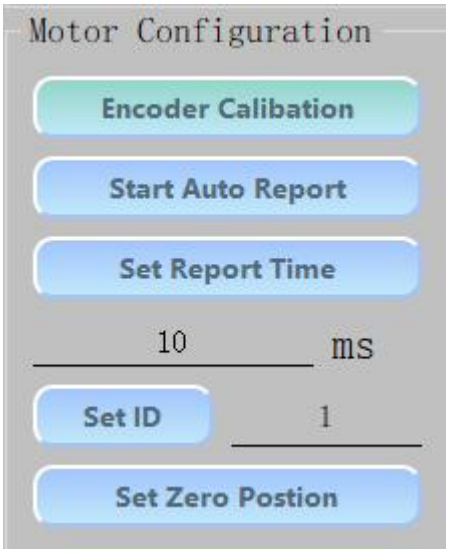
## 3.3. Motor settings

### 3.3.1. Motor connection settings



Connect the CAN to USB tool (install the ch340 driver, which works in AT mode by default), click Refresh Serial Port, open the serial port, and click Detect Device to detect the corresponding motor. The green text below is the motor type.

## 3.3.2. Motor configuration module




1. Recalibrate the motor magnetic encoder. Reinstalling the motor board and motor, or reconnecting the motor's three-phase wiring requires recalibrating the magnetic encoder.

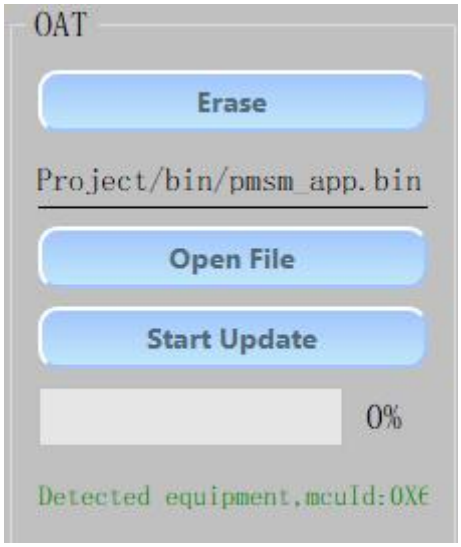
2. Enable active motor reporting. Click Start Reporting to enable active motor reporting in communication type 2. You can set the interval below, with a minimum of 10ms.

3. Set ID: Set the motor's CAN ID.

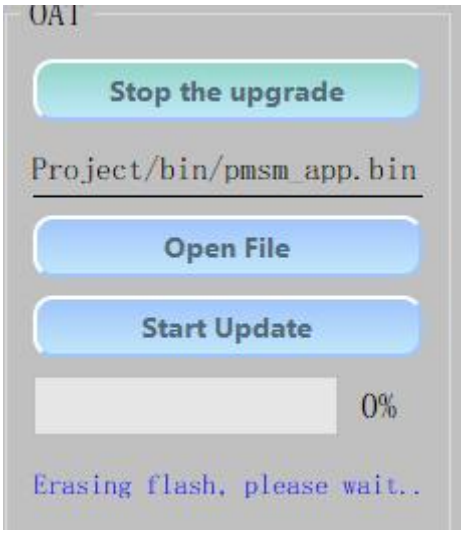
4. Set Zero Position: Set the current position to 0.

## 3.3.3. Motor upgrade module

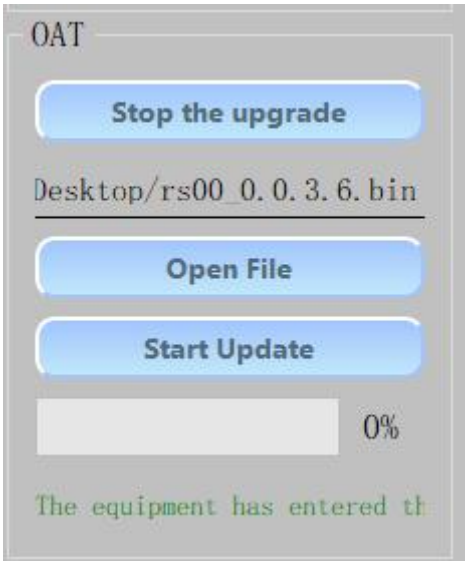
1. 1. Click to open the file and select the firmware to upgrade. The rs-0x in the firmware name is the selected motor type.



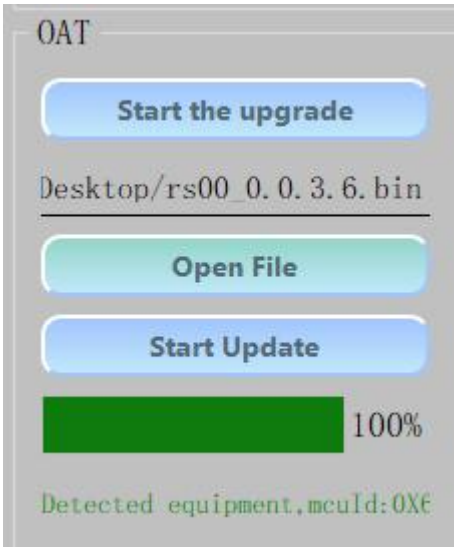

2. Click Start Upgrade, and the motor will enter the upgrade preparation stage.




3. When the green text "Device has entered upgrade mode" pops up, click to start the upgrade

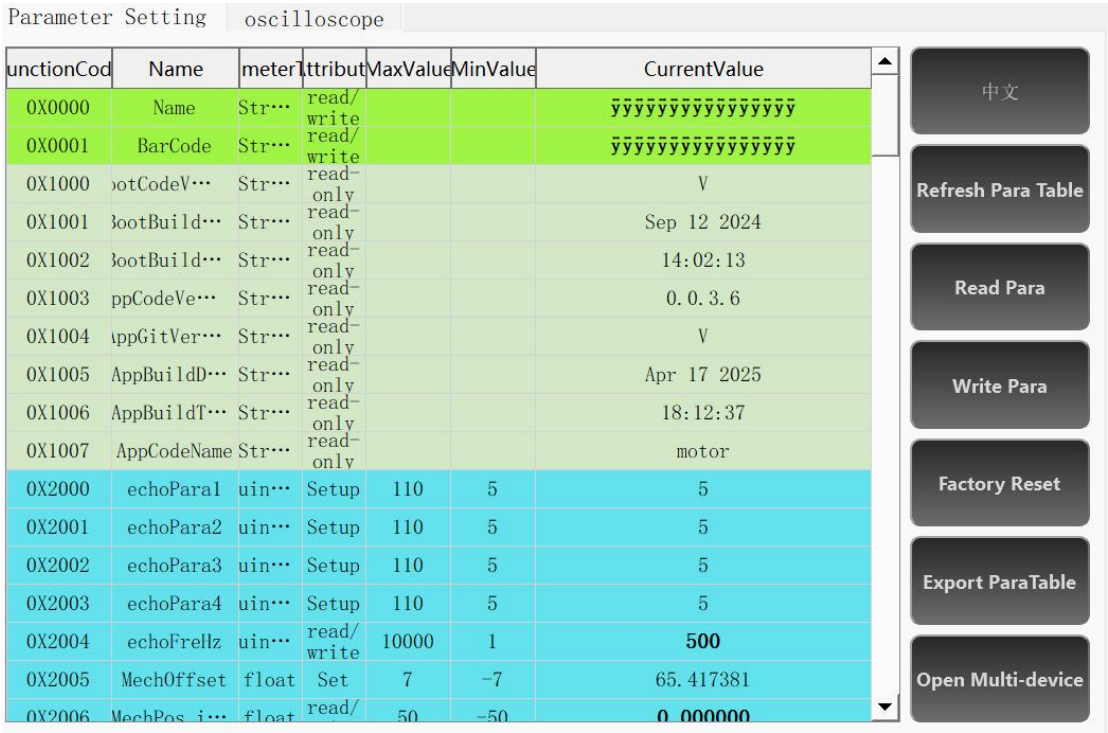



4. When the green text "Upgrade Successfully" pops up, the upgrade is complete.




If the green progress bar gets stuck halfway through the upgrade, you can click to stop the upgrade, or re-power on and re-enter the upgrade process. The internal program of the motor will not be lost after the upgrade fails. Please check whether the communication environment is good before upgrading again.

## 3.3.4. Parameter settings




After successfully connecting to the motor,

1. Click Refresh Parameter Table. "Updated Parameter Table Successfully" will appear at the top, indicating that the motor parameters have been successfully read (Note: The parameter table must be configured while the motor is in standby mode. If the motor is running, the parameter table refresh cannot be performed). The interface will display the motor's parameters. Parameters in blue are stored internally in the motor and can be modified in the Current Value field following the corresponding parameter.

2. Click Read Parameters to upload the motor parameters to the debugger. Parameters in light blue are observed parameters, which are collected and can be observed in real time.

3. Click Write Parameters to download the debugger parameters to the motor.

4. Click Restore Factory to restore the motor's default parameters for the latest firmware.

5. Click Export to export the current motor parameters in the parameter table.

6. Click Open Multi-Device Connection to connect the host computer to multiple motors. Note that because the parameter interfaces for different motor types vary, multi-device connection is only used for upgrades. After upgrading and debugging the motor, close multi-device connection and search for the motor again.

**Note: Please do not change the torque limit, protection temperature and overtemperature time of the motor. Our company will not bear any legal responsibility for any damage to human body or irreversible damage to joints caused by illegal operation of this product.**

| function code | name | parameter type | attribute | Maximum value | Minimum value | Current value (for reference) | 备注 |
|---|---|---|---|---|---|---|---|
| 0X0000 | Name | String | Read/Write | | | ÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿ | |
| 0X0001 | BarCode | String | Read/Write | | | ÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿÿ | |
| 0X1000 | BootCodeVersion | String | Read only | | | 0.1.5 | |
| 0X1001 | BootBuildDate | String | Read only | | | Mar 16 2022 | |
| 0X1002 | BootBuildTime | String | Read only | | | 20:22:09 | |
| 0X1003 | AppCodeVersion | String | Read only | | | 0.0.0.1 | Motor program version number |
| 0X1004 | AppGitVersion | String | Read only | | | 7b844b0fM | |
| 0X1005 | AppBuildDate | String | Read only | | | Apr 14 2022 | |
| 0X1006 | AppBuildTime | String | Read only | | | 20:30:22 | |
| 0X1007 | AppCodeName | String | Read only | | | Lingzu_motor | |
| 0X2000 | echoPara | uint16 | dispositio | 74 | 5 | 5 | |

| function code | name | parameter type | attribute | Maximum value | Minimum value | Current value (for reference) | 备注 |
|---|---|---|---|---|---|---|---|
| | 1 | | n | | | | |
| 0X2001 | echoPara2 | uint16 | disposition | 74 | 5 | 5 | |
| 0X2002 | echoPara3 | uint16 | disposition | 74 | 5 | 5 | |
| 0X2003 | echoPara4 | uint16 | disposition | 74 | 5 | 5 | |
| 0X2004 | echoFreHz | uint32 | Read/Write | 10000 | 1 | 500 | |
| 0X2005 | MechOffset | float | Settings | 7 | -7 | 4.619583 | Motor magnetic encoder Angle offset |
| 0X2006 | Chasu_offset | float | Read/Write | 50 | -50 | 4.52 | Reserved parameter |
| 0X2007 | Status1 | float | Read/Write | 17 | 0 | 17 | Torque limitation |
| 0X2008 | I_FW_MAX | float | Read/Write | 33 | 0 | 0 | Weak magnetic current value, default 0 |
| 0X2009 | CAN_ID | uint8 | Settings | 20 | 0 | 1 | Motor index, marking the joint position of the motor |
| 0X200a | CAN_MASTER | uint8 | Settings | 127 | 0 | 1 | id of this object |
| 0X200b | CAN_TIMEOUT | uint8 | Settings | 127 | 0 | 0 | can host id |
| 0X200c | status2 | uint32 | Read/Write | 100000 | 0 | 0 | can timeout threshold. The default value is 0 |
| 0X200d | status3 | int16 | Read/Write | 1500 | 0 | 800 | Reserved parameter |
| 0X200e | Status4 | uint32 | Read/Write | 1000000 | 1000 | 20000 | Reserved parameter |
| 0X200f | Status5 | float | Read/Write | 64 | 1 | 7.75 | Reserved parameter |
| 0X2010 | Status6 | uint8 | Read/Write | 1 | 0 | 1 | Reserved parameter |
| 0X2011 | cur_filt_gain | float | Read/Write | 1 | 0 | 0.9 | Current filtering parameter |
| 0X2012 | cur_kp | float | Read/Write | 200 | 0 | 0.025 | Current kp |
| 0X2013 | cur_ki | float | Read/Wri | 200 | 0 | 0.0258 | Current ki |

| function code | name | parameter type | attribute | Maximum value | Minimum value | Current value (for reference) | 备注 |
|---|---|---|---|---|---|---|---|
| | | | te | | | | |
| 0X2014 | spd_kp | float | Read/Write | 200 | 0 | 2 | Velocity kp |
| 0X2015 | spd_ki | float | Read/Write | 200 | 0 | 0.021 | Speed ki |
| 0X2016 | loc_kp | float | Read/Write | 200 | 0 | 30 | Position kp |
| 0X2017 | spd_filt_gain | float | Read/Write | 1 | 0 | 0.1 | Velocity filter parameter |
| 0X2018 | limit_spd | float | Read/Write | 200 | 0 | 2 | Location mode speed limit |
| 0X2019 | limit_cur | float | Read/Write | 23 | 0 | 23 | Position, Velocity mode current limit |
| 0X201a | limit_a | float | Read/Write | 100 | 0 | 0 | Reserved parameter |
| 0X201b | fault1 | float | Read/Write | 100 | 0 | 0 | Reserved parameter |
| 0X201c | fault2 | float | Read/Write | 27 | 0 | 0 | High speed segment offset |
| 0X201d | fault3 | float | Read/Write | 27 | 0 | 0 | The low end is offset |
| 0X201e | fault4 | float | Read/Write | 150 | 0 | | Velocity-mode acceleration |
| 0X201f | fault5 | float | Read/Write | 20 | 0 | | PP mode speed |
| 0X2020 | fault6 | float | Read/Write | 1000 | 0 | | PP mode acceleration |
| 0X2021 | fault7 | float | Read/Write | 100 | 0 | 0 | Zero marker |
| 0X2022 | baud | uint8 | Read/Write | 10 | 0 | 0 | Baud rate flag |
| 0X2023 | zero_sta | uint8 | Read/Write | | | 0 | Zero point flag |
| 0X2024 | position_offset | uint8 | Read/Write | 27 | 0 | 0 | Position offset |
| 0x2025 | protocol_1 | uint8 | Read/Write | | | 0 | Protocol flag |
| 0x2026 | damper | uint8 | Read/Write | 0 | 20 | 0 | Damping switch |

| function code | name | parameter type | attribute | Maximum value | Minimum value | Current value (for reference) | 备注 |
|---|---|---|---|---|---|---|---|
| 0x2027 | add_offset | float | Read/Write | -7 | 7 | 0 | Position offset parameter |
| 0X3000 | timeUse0 | uint16 | Read only | | | 5 | |
| 0X3001 | timeUse1 | uint16 | Read only | | | 0 | |
| 0X3002 | timeUse2 | uint16 | Read only | | | 10 | |
| 0X3003 | timeUse3 | uint16 | Read only | | | 0 | |
| 0X3004 | encoderRaw | int16 | Read only | | | 11396 | Magnetic encoder sampling value |
| 0X3005 | mcuTemp | int16 | Read only | | | 337 | mcu internal temperature, *10 |
| 0X3006 | motorTemp | int16 | Read only | | | 333 | Motor ntc temperature, *10 |
| 0X3007 | vBus(mv) | uint16 | Read only | | | 24195 | Bus voltage |
| 0X3008 | adc1Offset | int32 | Read only | | | 2084 | adc sampling channel 1 Zero current bias |
| 0X3009 | adc2Offset | int32 | Read only | | | 2084 | adc sampling channel 2 Zero current bias |
| 0X300a | adc1Raw | uint16 | Read only | | | 1232 | adc sampling value 1 |
| 0X300b | adc2Raw | uint16 | Read only | | | 1212 | adc sampling value 2 |
| 0X300c | VBUS | float | Read only | | | 36 | Bus voltage V |
| 0X300d | cmdId | float | Read only | | | 0 | id ring instruction, A |
| 0X300e | cmdIq | float | Read only | | | 0 | iq ring command, A |
| 0X300f | cmdlocref | float | Read only | | | 0 | Position loop command, rad |
| 0X3010 | cmdspdref | float | Read only | | | 0 | Speed loop command, rad/s |
| 0X3011 | cmdTorque | float | Read only | | | 0 | Torque instruction, nm |
| 0X3012 | cmdPos | float | Read only | | | 0 | mit Protocol Angle instruction |
| 0X3013 | cmdVel | float | Read only | | | 0 | mit Protocol Speed instruction |
| 0X3014 | rotation | int16 | Read only | | | 1 | Number of turns |
| 0X3015 | modPos | float | Read only | | | 4.363409 | Motor uncounted coil mechanical Angle, rad |
| 0X3016 | mechPos | float | Read only | | | 0.777679 | Load end loop mechanical Angle, rad |
| 0X3017 | mechVel | float | Read only | | | 0.036618 | Load speed: rad/s |

| function code | name | parameter type | attribute | Maximum value | Minimum value | Current value (for reference) | 备注 |
|---|---|---|---|---|---|---|---|
| 0X3018 | elecPos | float | Read only | | | 4.714761 | Electrical Angle |
| 0X3019 | ia | float | Read only | | | 0 | U-wire current, A |
| 0X301a | ib | float | Read only | | | 0 | V-wire current, A |
| 0X301b | ic | float | Read only | | | 0 | W-wire current, A |
| 0X301c | timeout | uint32 | Read only | | | 31600 | Timeout counter value |
| 0X301d | phaseOrder | uint8 | Read only | | | 0 | Directional marking |
| 0X301e | iqf | float | Read only | | | 0 | iq filter value，A |
| 0X301f | boardTemp | int16 | Read only | | | 359 | Plate temperature，*10 |
| 0X3020 | iq | float | Read only | | | 0 | iq Original value，A |
| 0X3021 | id | float | Read only | | | 0 | id Original value，A |
| 0X3022 | faultSta | uint32 | Read only | | | 0 | Fault status value |
| 0X3023 | warnSta | uint32 | Read only | | | 0 | Warning status value |
| 0X3024 | drv_fault | uint16 | Read only | | | 0 | The driver chip fault value is 1 |
| 0X3025 | drv_temp | int16 | Read only | | | 48 | The driver chip fault value is 2 |
| 0X3026 | Uq | float | Read only | | | 0 | Q-axis voltage |
| 0X3027 | as_angle | float | Read only | | | 0 | Magnetic encoder initial angle |
| 0X3028 | cs_angle | float | Read only | | | 0 | Differential magnetic encoder initial angle |
| 0X3029 | chasu_angle | float | Read only | | | 0 | Differential angle |
| 0X302a | v_bus | float | Read only | | | 0 | Motor voltage |
| 0X302b | ElecOffset | float | Read only | | | 24.195 | Electrical angle offset |
| 0X302c | torque_fdb | float | Read only | | | 0 | Torque feedback value，nm |
| 0X302d | rated_i | float | Read only | | | 8 | Rated current of motor |
| 0X302e | MechPos_init | float | Read only | | | 27 | Motor retention parameters |
| 0X302f | instep | float | Read only | | | 0 | Motor protection parameters |
| 0X3030 | status | uint8 | Read only | | | 0 | Reserved parameters |
| 0X3031 | cmdlocre | float | Read only | | | 0 | Position setpoint |

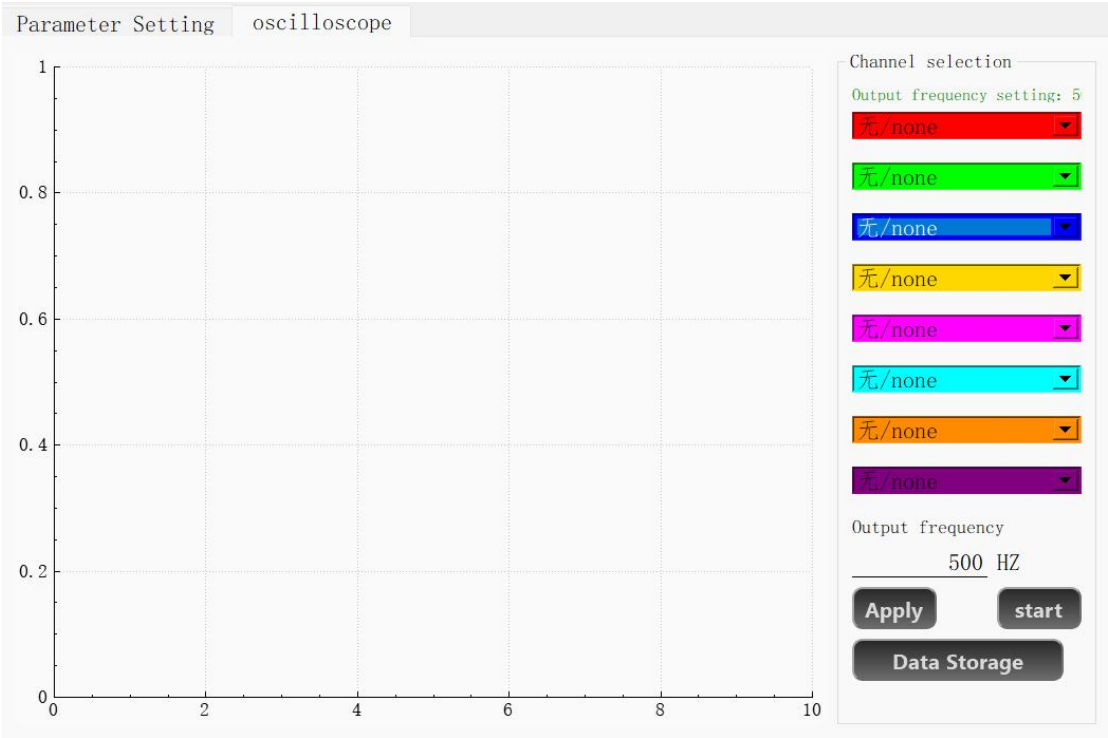
| function code | name | parameter type | attribute | Maximum value | Minimum value | Current value (for reference) | 备注 |
|---|---|---|---|---|---|---|---|
| | f | | | | | | |
| 0X3032 | vel_max | float | Read only | | | 0 | Motor speed setpoint |
| 0X3033 | fault1 | float | Read only | | | 0 | Motor position determination parameters |
| 0X3034 | fault2 | float | Read only | | | 0 | Motor position determination parameters |
| 0X3035 | fault3 | float | Read only | | | 0 | Motor position determination parameters |
| 0X3036 | fault4 | float | Read only | | | 0 | Motor position determination parameters |
| 0X3037 | fault5 | float | Read only | | | 0 | Log failure |
| 0X3038 | fault6 | uint32 | Read only | | | 0 | Log failure |
| 0X3039 | fault7 | uint32 | Read only | | | 0 | Log failure |
| 0X303a | fault8 | uint32 | Read only | | | 0 | Log failure |
| 0X303b | mcOverTemp | int16 | Read only | | | 0 | Over-temperature threshold |
| 0X303c | Kt_Nm/Amp | float | Read only | | | 0 | Torque coefficient |
| 0X303d | Tqcali_Type | uint8 | Read only | | | 0 | Motor type |
| 0X303e | theta_mech_1 | float | Read only | | | 0 | Type 2 low-speed angle |
| 0X303f | adc0ffset_1 | uint32 | Read only | | | 2053 | ADC sampling channel 1 zero current offset |
| 0X3040 | adc0ffset_2 | uint32 | Read only | | | 2057 | ADC sampling channel 2 zero current offset |
| 0X3041 | can_status | uint8 | Read only | | | 1 | CAN status |
| 0X3042 | position | float | Read only | | | 16.558939 | Initialization position |
| 0X3043 | chasu_an | float | Read only | | | 2.106539 | Motor position |

| function code | name | parameter type | attribute | Maximum value | Minimum value | Current value (for reference) | 备注 |
|---|---|---|---|---|---|---|---|
| | gle_init | | | | | | determination parameters |
| 0X3044 | chasu_angle_out | float | Read only | | | 18.958853 | Motor position determination parameters |
| 0X3045 | motormechinit | float | Read only | | | 16.558939 | Motor position determination parameters |
| 0X3046 | mech_angle_init2 | float | Read only | | | 1.839882 | Motor position determination parameters |
| 0X3047 | mech_angle_rotat | int16 | Read only | | | 0 | Motor position determination parameters |
| 0X3048 | coder_reg | uint16 | Read only | | | 1 | System parameters |
| 0X3049 | pos_cnt1 | uint16 | Read only | | | 0 | System parameters |

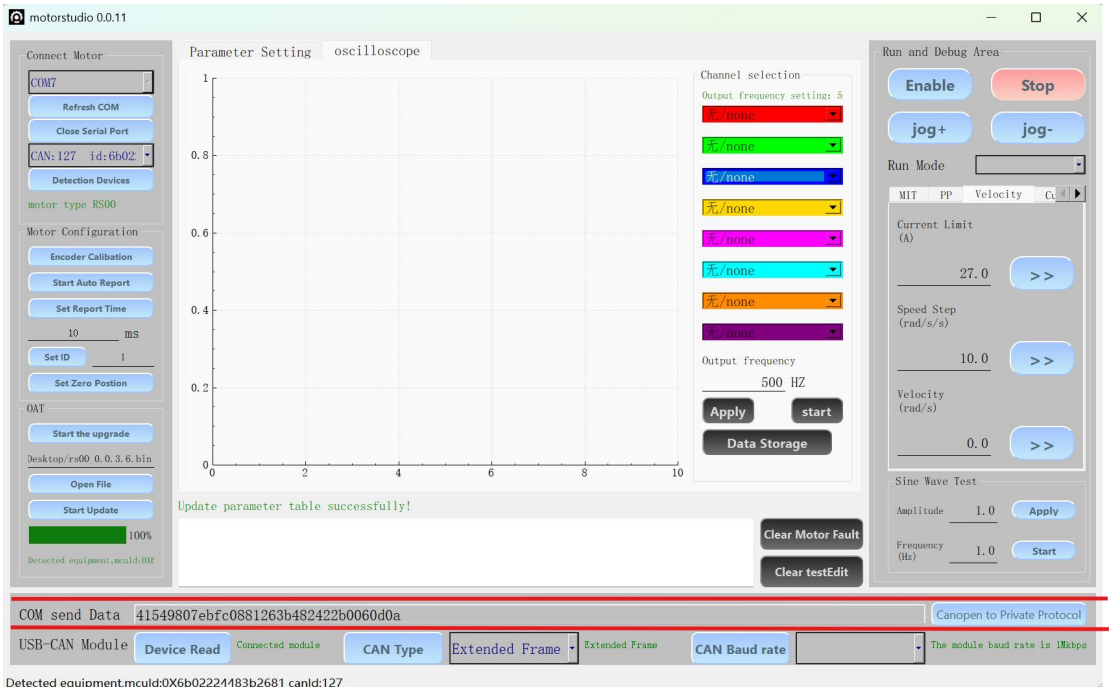
## 3.3.5. Oscilloscope

The interface supports viewing and observing the graph generated by real-time data, including motor Id/Iq current, temperature, real-time speed at the output end, rotor (encoder) position, output end position, etc.

Click on the oscilloscope module in the analysis module, select the appropriate parameters in the channel (parameter meaning can be referred to the parameter table), set the output frequency, click on the start plot to observe the data graph, stop the plot to stop the observation graph.




The command sent is in the communication command box below




Communication box instruction example:

41 54 90 07 e8 0c 08 05 70 00 00 01 00 00 00 0d 0a

The meaning is as follows

| | | | | |
|---|---|---|---|---|
| **41 54** | **90 07 e8 0c** | **8** | **05 70 00 00 01 00 00** | **0d 0a** |

| | | | | |
|---|---|---|---|---|
| | | | **00** | |
| frame header | Number of data bits | extended frame | data frame | frame tail |

The translation of extended frame canid into real canid requires the following transformations:

90 07 e8 0c converts to binary as 1001 0000 0000 0111 1110 1000 0000 1100, remove the 100 on the right and it becomes 1 0010 0000 0000 1111 1101 0000 0001, convert it to hexadecimal, It is 12 00 FD 01. According to the communication protocol, the meaning is as follows:

| **12 in hexadecimal** | **0** | **FD** | **1** |
|---|---|---|---|
| Communication type 18 (in decimal base) | No meaning | host id | motor canid |

## 3.3.6. can communication failure protection

When the value of CAN_TIMEOUT is 0, this function is disabled

When the CAN_TIMEOUT value is non-0, when the motor does not receive the can command within a certain period of time, the motor enters the reset mode, and 20000 is 1s

## 3.3.7. Motor fault instructions

Function code 0x3022 indicates the fault code, where

bit14:motor blocking overload algorithm protection

bit7: Encoder uncalibrated: Motor uncalibrated encoder

bit3: Overvoltage fault: the motor voltage exceeds the protection voltage by 60V

bit2: Undervoltage fault: the motor voltage is lower than the protection voltage of 12V

bit1: Driver chip failure: Motor driver chip failure reported

bit0: Motor overtemperature fault: motor thermistor temperature exceeds 145

degrees

Function code 0x3024 is driver chip fault code 1. The specific faults are as follows

**Table 11. Fault Status Register 1 Field Descriptions**

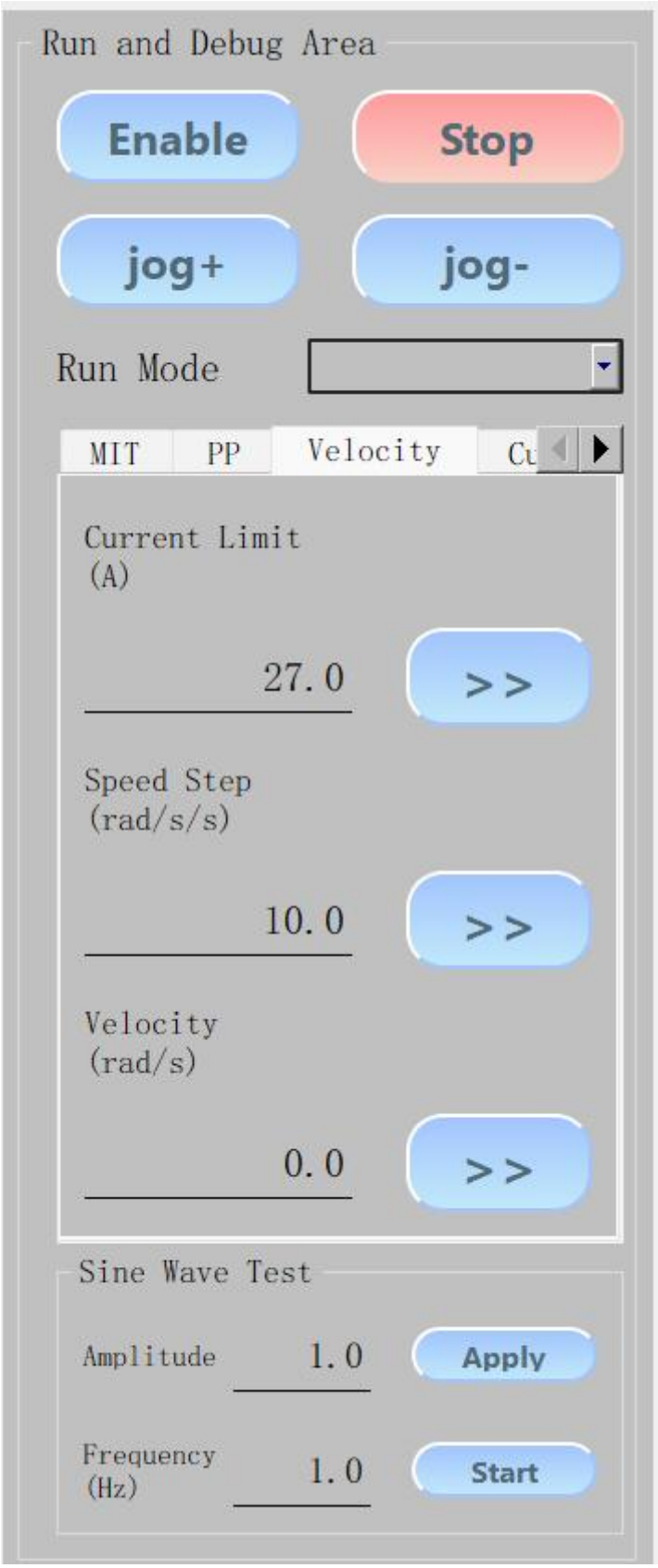
| Bit | Field | Type | Default | Description |
|---|---|---|---|---|
| 10 | FAULT | R | 0b | Logic OR of FAULT status registers. Mirrors nFAULT pin. |
| 9 | VDS_OCP | R | 0b | Indicates VDS monitor overcurrent fault condition |
| 8 | GDF | R | 0b | Indicates gate drive fault condition |
| 7 | UVLO | R | 0b | Indicates undervoltage lockout fault condition |
| 6 | OTSD | R | 0b | Indicates overtemperature shutdown |
| 5 | VDS_HA | R | 0b | Indicates VDS overcurrent fault on the A high-side MOSFET |
| 4 | VDS_LA | R | 0b | Indicates VDS overcurrent fault on the A low-side MOSFET |
| 3 | VDS_HB | R | 0b | Indicates VDS overcurrent fault on the B high-side MOSFET |
| 2 | VDS_LB | R | 0b | Indicates VDS overcurrent fault on the B low-side MOSFET |
| 1 | VDS_HC | R | 0b | Indicates VDS overcurrent fault on the C high-side MOSFET |
| 0 | VDS_LC | R | 0b | Indicates VDS overcurrent fault on the C low-side MOSFET |

Function code 0x3025 is driver chip fault code 2. The specific faults are as follows

**Table 12. Fault Status Register 2 Field Descriptions**

| Bit | Field | Type | Default | Description |
|---|---|---|---|---|
| 10 | SA_OC | R | 0b | Indicates overcurrent on phase A sense amplifier (DRV8353xS) |
| 9 | SB_OC | R | 0b | Indicates overcurrent on phase B sense amplifier (DRV8353xS) |
| 8 | SC_OC | R | 0b | Indicates overcurrent on phase C sense amplifier (DRV8353xS) |
| 7 | OTW | R | 0b | Indicates overtemperature warning |
| 6 | GDUV | R | 0b | Indicates VCP charge pump and/or VGLS undervoltage fault condition |
| 5 | VGS_HA | R | 0b | Indicates gate drive fault on the A high-side MOSFET |
| 4 | VGS_LA | R | 0b | Indicates gate drive fault on the A low-side MOSFET |
| 3 | VGS_HB | R | 0b | Indicates gate drive fault on the B high-side MOSFET |
| 2 | VGS_LB | R | 0b | Indicates gate drive fault on the B low-side MOSFET |
| 1 | VGS_HC | R | 0b | Indicates gate drive fault on the C high-side MOSFET |
| 0 | VGS_LC | R | 0b | Indicates gate drive fault on the C low-side MOSFET |

## 3.4. Control Demo



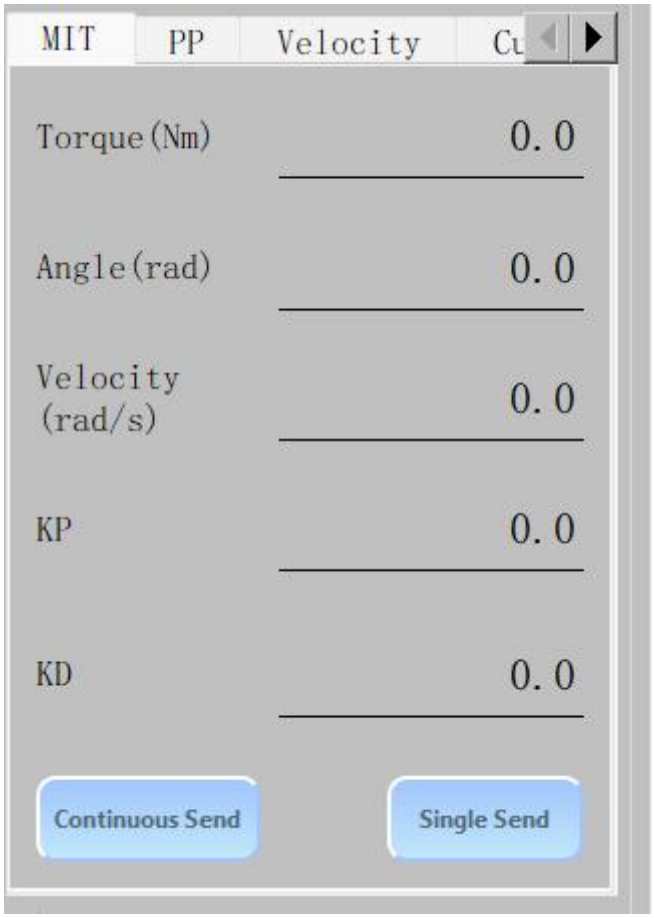
Jog Run

Click JOG +/- to run the motor forward and reverse at a speed of 1 rad/s.

Control Mode Switching

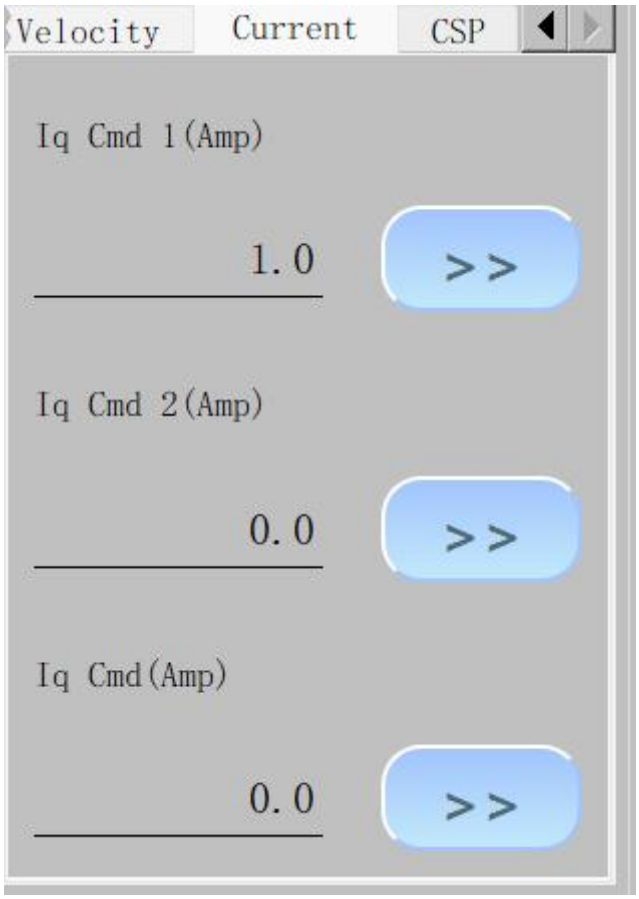
Select the desired control mode in the command box to the right of the run mode.

## 3.4.1. Operation and control mode

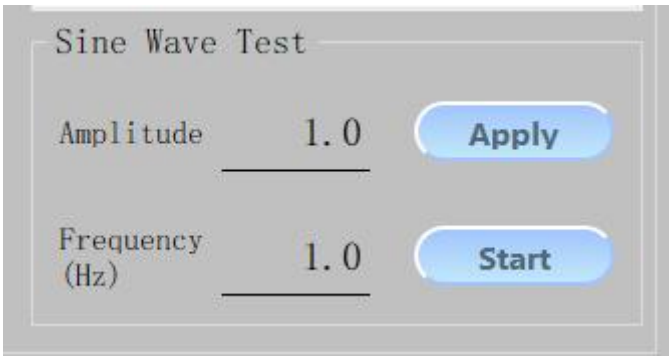



1. Switch the control mode to operation mode.
2. The motor starts running and enters motor_mode.
3. Set five parameter values and click Start or Send continuously. The motor will return feedback frames and run according to the target command.
4. Click Stop to stop the motor and terminate the continuous sending of commands.

## 3.4.2. Current mode




1. Switch the control mode to current mode.
2. The motor starts running and enters motor_mode.
3. Set the current command value for Iq Command 1 (A). Click the >> button on the right. The motor will follow the current command.
4. Click Stop to stop the motor.




Motor Current Sine Test

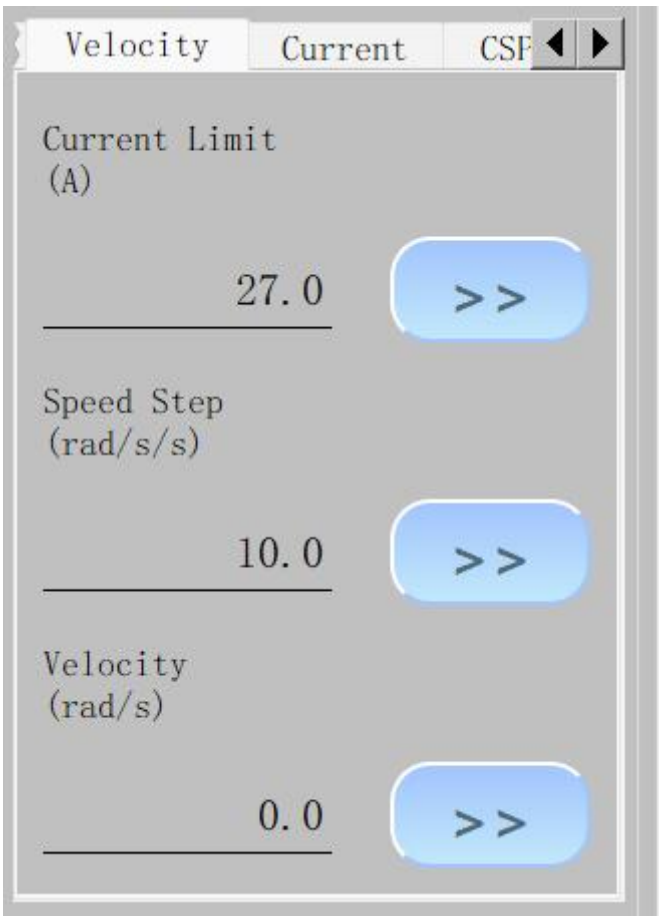
1. Switch the control mode to current mode.

2. The motor starts running and enters motor_mode.

3. Set the amplitude and frequency, click OK, and then click Start. The corresponding mode target command will be planned according to the sine law.

4. Click Stop to stop the motor.

## 3.4.3. Speed Mode



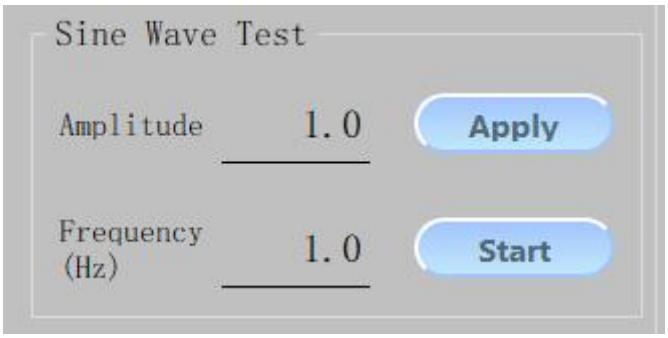

1. Switch the control mode to speed mode.

2. The motor starts running and enters motor_mode.

3. First, set the current limit (maximum phase current) and speed step value (motor acceleration). If no settings are made, the motor will operate at the default values. Finally, set the speed command (target speed). The motor will

follow the command.

4. Click Stop to stop the motor.




Motor Speed Sine Test

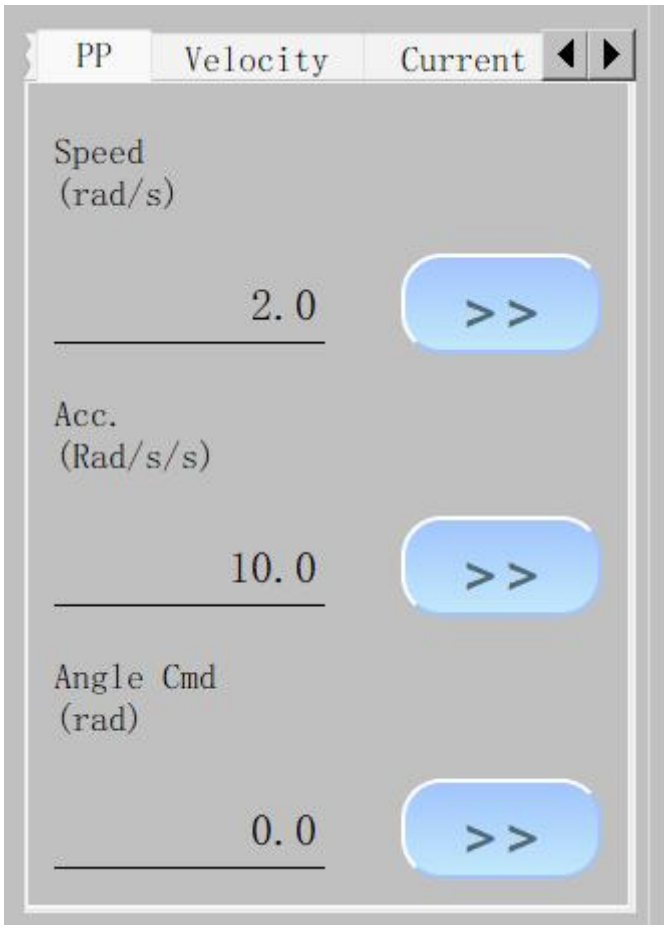
1. Switch the control mode to speed mode.

2. The motor starts running and enters motor_mode.

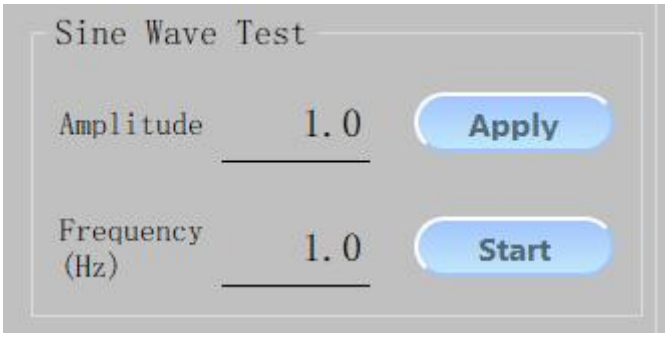
3. Set the amplitude and frequency, click OK, and then click Start. The corresponding mode target command will be planned according to the sine law.

4. Click Stop to stop the motor.

### 3.4.4. Position Mode (PP)



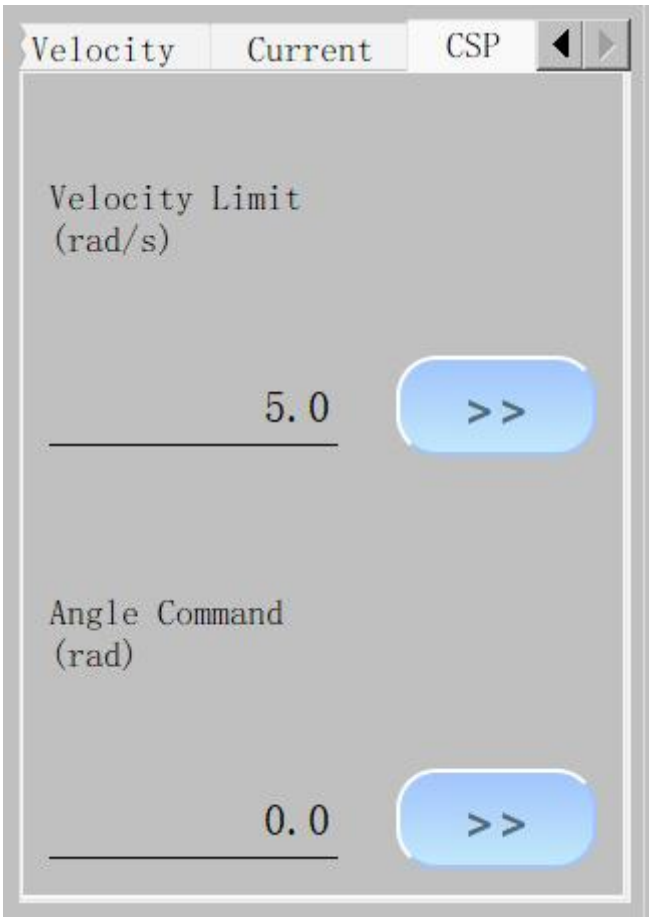

1. Switch the control mode to interpolation position mode.
2. Start the motor and enter motor_mode.
3. First, set the speed and acceleration. If not set, the motor will operate at the default values. Finally, set the position command (target position). The motor will follow the command.
4. Set the speed to 0 to stop the motor at the current position. To continue operation, re-issue the speed and position.
5. Click Stop to stop the motor.



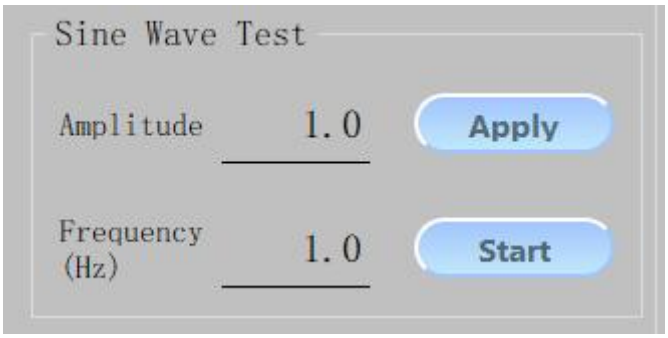

Motor Position Sine Test

1. Switch the control mode to interpolation position mode.
2. Start the motor and enter motor_mode.
3. Set the amplitude and frequency, click OK, and then click Start. The corresponding mode target command will be planned according to the sine law.
4. Click Stop to stop the motor.

### 3.4.5. Position Mode（CSP）




1. Switch the control mode to position mode.
2. The motor starts running and enters motor_mode.
3. Set the speed first. If no speed setting is made, the motor will run at the default value. Finally, set the position command (target position). The motor will follow the command.
4. Click Stop to stop the motor.



Motor position sinusoidal test

1. Switch the control mode to position mode.
2. The motor starts running and enters motor_mode.
3. Set the amplitude and frequency, click OK, and then click Start. The corresponding mode target command is then sinusoidally planned.
4. Click Stop to stop the motor.

# 4. Driver protocol and instructions

The motor communication is the CAN 2.0 communication interface, the baud rate is 1Mbps, and the extended frame format is adopted as follows:

| data field | 29-bit ID | | | 8Byte data field |
|---|---|---|---|---|
| Size | Bit28~bit24 | bit23~8 | bit7~0 | Byte0~Byte7 |
| Description | Communication type | data area 2 | Destination address | data area 1 |

The control modes supported by the motor include:

- Operation control mode: set 5 parameters of motor operation control;
- Current mode: the specified Iq current of the given motor;
- Velocity mode: the specified running speed of the given motor;
- Position mode: Given the specified position of the motor, the motor will run to the specified position;

## 4.1. Description of the communication protocol type

### 4.1.1. Communication type 0: Get device ID

Gets the device's ID and 64-bit MCU unique identifier

| data field | 29-bit ID | | | 8Byte data field |
|---|---|---|---|---|
| Size | Bit28~bit24 | bit23~8 | bit7~0 | Byte0~Byte7 |
| Description | 0x0 | bit15~8: identifies host CAN_ID | target motor CAN_ID | 0 |

Reply frame:

| data field | 29-bit ID | | | 8Byte data field |
|---|---|---|---|---|
| Size | Bit28~bit24 | bit23~8 | bit7~0 | Byte0~Byte7 |
| Description | 0x0 | target motor CAN_ID | 0XFE | 64-bit MCU unique identifier |

## 4.1.2. Communication Type 1: operation control mode motor control instruction

| Data field | 29 bit ID | | | 8Byte data field |
|---|---|---|---|---|
| Size | Bit28~bit24 | bit23~8 | bit7~0 | Byte0~Byte7 |
| description | 0x1 | Byte2: Torque (0~65535) corresponds to (-60Nm~60Nm) | target motor CAN _ID | Byte0~1: target Angle [0~65535] corresponds to (-4π~4π) Byte2~3: Target angular velocity [0~65535] corresponds to (-20rad/s~20rad/s) Byte4~5: Kp [0~65535] corresponds to (0.0~5000.0) Byte6~7: Kd [0 to 65535] corresponds to the above data (0.0 to 100.0). After the conversion, the high byte is in front and the low byte is in |

Response frame: Response motor feedback frame (see communication type 2)

Communication Type 2: motor feedback data

| Data field | 29 bit ID | | | 8Byte data field |
|---|---|---|---|---|
| Size | Bit28~bit24 | bit23~8 | bit7~0 | Byte0~Byte7 |
| Description | 0x2 | Bit8~Bit15: CAN ID of the current motor bit21~16: fault information (0 none 1 has) bit21: uncalibrated bit20: Uncalibrated bit20: Gridlock overload fault bit19: magnetic coding fault bit18: overtemperature bit17: Three-phase overcurrent fault bit16: undervoltage fault bit22~23: Mode status 0: Reset mode [reset] 1: Cali mode [calibration] 2: Motor mode [Run] | host CAN _ID | Byte0~1: The current Angle [0~65535] Corresponding to (-4π~4π) Byte2~3: Current angular velocity [0~65535] corresponds to (-20rad/s~20rad/s) Byte4~5: Current torque [0~65535] corresponds to (-60Nm~60Nm) Byte6~7: Current temperature: Temp(Celsius) *10 If the value is higher than 10, the high byte is first and the low byte is last |

### 4.1.3. Communication Type 3: Motor enabled to run

| data field | 29-bit ID | | | 8Byte data field |
|---|---|---|---|---|
| Size | Bit28~bit24 | bit23~8 | bit7~0 | Byte0~Byte7 |
| Description | 0x3 | bit15~8: identifies the main CAN_ID | and target motor CAN_ID | |

Response frame: Response motor feedback frame (see communication type 2)

### 4.1.4. Communication Type 4: Motor stops running

| data field | 29-bit ID | | | 8Byte data field |
|---|---|---|---|---|
| Size | Bit28~bit24 | bit23~8 | bit7~0 | Byte0~Byte7 |
| description | 0x4 | bit15~8: used to identify the main CAN_ID | target motor CAN_ID | When the motor is running normally, 0 must be cleared in the data field. Byte[0]=1: The fault is cleared. |

Response frame: Response motor feedback frame (see communication type 2)

## Version number read frame

| Data field | 29 bit ID | | | 8Byte data field |
|---|---|---|---|---|
| Size | Bit28~bit24 | bit23~8 | bit7~0 | Byte0~Byte7 |
| description | 0x4 | bit15~8: used to identify the main CAN_ID | target motor CAN_ID | Byte[0]=0x00<br>Byte[1]=0xC4 |

Response frame:

| data field | 29 位 ID | | | 8Byte data field |
|---|---|---|---|---|
| Size | Bit28~bit24 | bit23~8 | bit7~0 | Byte0~Byte7 |
| description | 0x2 | Bit8~Bit15: CAN ID of the current motor bit21~16: fault information (0 none 1 has) bit21: uncalibrated bit20: Uncalibrated bit20: Gridlock overload fault bit19: magnetic coding fault bit18:overtemperature | target motor CAN_ID | Byte0=0x00<br>Byte1=0xC4<br>Byte2=0x56<br>Byte3~6:Motor version number Order from high to low |

| | | | | |
|---|---|---|---|---|
| | | bit17: Three-phase overcurrent fault bit16: undervoltage fault bit22~23: Mode status 0: Reset mode [reset] 1: Cali mode [calibration] 2: Motor mode [Run] | | |

### 4.1.5. Communication type 6: Set motor mechanical zero

| data field | 29-bit ID | | | 8Byte data field |
|---|---|---|---|---|
| Size | Bit28~bit24 | bit23~8 | bit7~0 | Byte0~Byte7 |
| description | 0x6 | bit15~8: Identifies the main CAN_ID | and target motor CAN_ID | Byte[0]=1 |

Response frame: Response motor feedback frame (see communication type 2)

Communication type 7: Set motor CAN_ID

Change the current motor CAN_ID, effective immediately.

| data field | 29-bit ID | | | 8Byte data field |
|---|---|---|---|---|
| Size | Bit28~bit24 | bit23~8 | bit7~0 | Byte0~Byte7 |
| description | 0x7 | bit15~8: used to identify main CAN_ID Bit16~23: preset CAN_ID | Target motor CAN_ID | |

Answer frame: Answer motor broadcast frame (see communication type 0)

### 4.1.6. Communication type 17: Single parameter read

| Data field | 29 bit ID | | | 8Byte data field |
|---|---|---|---|---|
| Size | Bit28~bit24 | bit23~8 | bit7~0 | Byte0~Byte7 |
| description | 0x11 | bit15~8: Used to identify the main CAN_ID | target motor CAN_ID | Byte0~1: index. For details, see the readability parameter table below Byte2~3:00 Byte4~7: In data above 00, the low byte is first and the high byte is second ( |

Reply frame:

| Data field | 29 bit ID | | | 8Byte data field |
|---|---|---|---|---|
| Size | Bit28~bit24 | bit23~8 | bit7~0 | Byte0~Byte7 |
| description | 0x11 | bit15~8: indicates that the master CAN_ID Bit23~16:00 indicates that the master CAN_ID is successfully read. 01 indicates that the master can_ID | Byte0~1: Byte2~3:00 Byte4~7:Parameter data. 1 byte of data above Byte4 is preceded by low bytes and followed by high bytes at | |

## 4.1.7. Communication type 18: Single parameter write (lost in power failure)

With type 22, the parameter starting with function code 0x20 of the parameter table in the upper computer module can be saved

| Data field | 29 bit ID | | | 8Byte data field |
|---|---|---|---|---|
| Size | Bit28~bit24 | bit23~8 | bit7~0 | Byte0~Byte7 |
| description | 0x12 | bit15~8: Used to identify the main CAN_ID | target motor CAN_ID | Byte0~1: index. For details, see the readability parameter table below Byte2~3: 00 Byte4~7: Parameter data In the preceding data, the low byte is in the front and the high byte is in the rear |

Response frame: Response motor feedback frame (see communication type 2)

## 4.1.8. Communication type 21: Fault feedback frame

| data field | 29-bit ID | | | 8Byte data field |
|---|---|---|---|---|
| Size | Bit28~bit24 | bit23~8 | bit7~0 | Byte0~Byte7 |
| description | 0x15 | bit15~8: motor | identifies the main | Byte0~3: fault value (non-0: faulty; 0: faulty). Normal) |

| | | | | |
|---|---|---|---|---|
| | | CAN_ID | CAN_ID | Bit 16: A-phase current sampling overcurrent<br>Bit 14: Motor stall overload algorithm protection<br>Bit 9: Position initialization fault<br>Bit 8: Hardware identification fault<br>Bit 7: Encoder uncalibrated: Motor encoder not calibrated<br>Bit 5: C-phase current sampling overcurrent<br>Bit 4: B-phase current sampling overcurrent<br>bit3: overvoltage fault<br>bit2: undervoltage fault<br>bit1: driver chip fault bit0: motor overtemperature fault, Default 145 ° C<br>Byte4~7: warning Value<br>bit0: motor overtemperature warning, the default is 135 ° c |

## 4.1.9. Communication type 22: Motor data save frame

| data field | 29-bit ID | | | 8Byte data field |
|---|---|---|---|---|
| Size | Bit28~bit24 | bit23~8 | bit7~0 | Byte0~Byte7 |
| Description | 0x16 | bit15~8: identifies the main CAN_ID | and target motor CAN_ID | 01 02 03 04 05 06 07 08 |

Response frame: Response motor feedback frame (see communication type 2)

## 4.1.10. Communication type 23: Motor baud rate modification frame (re-power-on effect)

| Data field | 29 bit ID | | | 8Byte data field |
|---|---|---|---|---|
| Size | Bit28~bit24 | bit23~8 | bit7~0 | Byte0~Byte7 |
| description | 0x17 | bit15~8: used to identify the main CAN_ID | target motor CAN_ID | 01 02 03 04 05 06 F_CMD<br>Among them, the F_CMD byte is the motor baud rate Among them, 01 is 1M 02 is 500K 03 is 250K 04 is 125K |

Response frame: Response motor feedback frame (see communication type 0)

### 4.1.11. Communication type 24: The motor actively reports frames

| data field | 29-bit ID | | | 8Byte data field |
|---|---|---|---|---|
| Size | Bit28~bit24 | bit23~8 | bit7~0 | Byte0~Byte7 |
| description | 0x18 | bit15~8: identifies the main CAN_ID | target motor CAN_ID | 01 02 03 04 05 06 F_CMD<br>Among them, the F_CMD byte is the motor reporting switch 00 is to disable active reporting (default) 01 To enable active reporting, the default reporting interval is 10ms |

Response frame:

| data field | 29 位 ID | | | 8Byte data field |
|---|---|---|---|---|
| Size | Bit28~bit24 | bit23~8 | bit7~0 | Byte0~Byte7 |
| description | 0x18 | Bit8~Bit15: CAN ID of the current motor bit21~16: fault information (0 none 1 has) bit21: uncalibrated bit20: Uncalibrated bit20: Gridlock overload fault bit19: magnetic coding fault bit18:overtemperature bit17: Three-phase overcurrent fault bit16: undervoltage fault bit22~23: Mode status 0: Reset mode [reset] 1: Cali mode [calibration] 2: Motor mode [Run] | target motor CAN_ID | Byte0~1: The current Angle [0~65535] Corresponding to (-4π ~4π) Byte2~3: Current angular velocity [0~65535] corresponds to (-20rad/s~20rad/s) Byte4~5: Current torque [0~65535] corresponds to (-60Nm~60Nm) Byte6~7: Current temperature: Temp(Celsius) *10 If the value is higher than 10, the high byte is first and the low byte is last |

### 4.1.12. Communication type 25: Motor protocol modification frame (re-power-on effect)

| Data field | 29 bit ID | | | 8Byte data field |
|---|---|---|---|---|
| Size | Bit28~bit24 | bit23~8 | bit7~0 | Byte0~Byte7 |
| description | 0x19 | bit15~8: used to identify the main CAN_ID | target motor CAN_ID | 01 02 03 04 05 06 F_CMD<br>Among them, the F_CMD byte is the motor protocol type Among them, 0 is a private protocol (default)<br>1 is the Canopen protocol<br>2 is the MIT protocol |

Response frame: Response motor feedback frame (see communication type 0)

## 4.1.13. Read and write a single parameter list

| index | | Description | Type | Number of bytes | | R/W Read and write permission |
|---|---|---|---|---|---|---|
| 0X7005 | run_mode | 0: operation mode<br>1: position mode (PP)<br>2: Velocity mode<br>3: Operation mode<br>Current mode<br>5: Position mode (CSP) | uint8 | 1 | | W/R |
| 0X7006 | iq_ref | Current mode Iq command | float | 4 | -43 to 43A | W/R |
| 0X700A | spd_ref | Rotational Velocity mode Rotational speed command | float | 4 | -20to 20rad/s | W/R |
| 0X700B | limit_torque | torque limit | float | 4 | 0 to 60Nm | W/R |
| 0X7010 | cur_kp | Kp | float | 4 | The default value is 0.17 | W/R |
| 0X7011 | cur_ki | Ki | float | 4 | The default value is 0.012 | W/R |
| 0X7014 | cur_filt_gain | filt_gain | float | 4 | 0 to 1.0, The default value is 0.1 | W/R |
| 0X7016 | loc_ref | Position Mode Angle instruction | float | 4 | rad | W/R |
| 0X7017 | limit_spd | Location mode (CSP) speed limit | float | 4 | 0 to 20rad/s | W/R |
| 0X7018 | limit_cur | Velocity position mode Current limitation | float | 4 | 0 to 43A | W/R |
| 0x7019 | mechPos | Mechanical Angle of the loading coil | float | 4 | rad | R |
| 0x701A | iqf | iq Filter | float | 4 | -43 to 43A | R |
| 0x701B | mechVel | Speed of the load | float | 4 | -20 to | R |

| index | | Description | Type | Number of bytes | | R/W Read and write permission |
|---|---|---|---|---|---|---|
| | | | | | 20rad/s | |
| 0x701C | VBUS | Bus voltage | float | 4 | V | R |
| 0x701E | loc_kp kp | at | float | 4 | The default value is 60 | W/R |
| 0x701F | spd_kp | Indicates the speed kp | float | 4 | The default value is 6 | W/R |
| 0x7020 | spd_ki | ki | float | 4 | The default value is 0.02 | W/R |
| 0x7021 | spd_filt_gain | Speed filter value | float | 4 | The default value is 0.1 | W/R |
| 0x7022 | acc_rad | velocity mode acceleration | float | 4 | The default value is 20rad/s^2 | W/R |
| 0x7024 | vel_max | Location mode (PP) speed | float | 4 | The default value is 10rad/s | W/R |
| 0x7025 | acc_set | Location mode (PP) acceleration | float | 4 | The default value is 10rad/s^2 | W/R |
| 0x7026 | EPScan_time | Indicates the report time. 1 indicates 10ms. Plus 1 increments by 5ms | uint16 | 2 | The default value is 1 | W/R |
| 0x7028 | canTimeout | can The timeout threshold, 20000 is 1s | uint32 | 4 | The default value is 0 | W/R |
| 0x7029 | zero_sta | Indicates the zero flag bit, 0 means 0-2π and 1 means -π-π | uint8 | 1 | The default is 0 | W/R |
| 0x702A | damper | Damping switch | uint8 | 1 | The default value is 0 | W/R |
| 0x702B | add_offset | Zero offset | float | 4 | The default is 0 | W/R |
| 0x702C | alveolous_open | Cogging Compensation Switch | uint8 | 1 | The default is 0 | W/R |

| index | | Description | Type | Number of bytes | | R/W Read and write permission |
|---|---|---|---|---|---|---|
| 0x702D | iq_test | Motor Initialization Calibration Switch | uint8 | 1 | The default is 0 | W/R |
| 0x702E | dcc_set | Deceleration in Motor PP Mode (Normally Disabled) | float | 4 | The default is 10rad/s^2 | W/R |

### 4.1.14. Read example:

Take reading loc_kp as an example:

Read instruction is

| Size | Bit28~bit24 | bit23~8 | bit7~0 | Byte0~Byte7 |
|---|---|---|---|---|
| | 0x11 | 0x00FD | 0x7F | 1E 70 00 00 00 00 00 00 |
| Description | Type 17 | Host id 0xFD | Target motor CAN_ID 7F | Byte0~1: index, corresponding to loc_kp |

The feedback instruction is

| Size | Bit28~bit24 | bit23~8 | bit7~0 | Byte0~Byte7 |
|---|---|---|---|---|
| | 0x11 | 0x007F | 0xFD | 1E 70 00 00 00 00 F0 41 |
| Description | Type 17 | bit15~8: Target motor CAN_ID 7F | Host id 0xFD | Byte0~1: index, corresponding to loc_kp Byte4~7:loc_kp value 30, high right byte, (32-bit single precision) hexadecimal IEEE-754 standard floating point number |

## 4.2. Motor Function Description

*(If the following features are unavailable, please upgrade to the latest version via the official Git repository.)*

---

### 4.2.1. Active Reporting

- **Disabled by default**. Enable via **Type 24**.

- Report type: **Type 2** (default interval: **10ms**). Adjust interval by modifying `EPScan_time` via **Type 18**.

---

### 4.2.2. Zero-Point Flag (`zero_sta`)

- Modify via:
  - **Host computer**
  - **Type 18** (*requires saving via **Type 22** for communication*)
- **Default flag**: `0` → Power-on position range: **0–2π**.
- **If set to `1`**: Power-on position range: **π–π**.

---

### 4.2.3. Type 2 Update

- Updated to **periodic looping within -4π–4π** (enables cycle counting).
- **Note**: Position interface parameters must be adjusted:
  - `P_MIN`: **12.57f**
  - `P_MAX`: **12.57f**

### 4.2.4. Protocol Switching *(Requires CAN adapter)*

- Methods:
  - Modify `protocol_1` via host computer.
  - Send **Type 25** command.
- **Reboot required** after switching.
- **Post-switch CAN commands**:
  - **CANopen**: Send extended frame (protocol switch frame).
  - **MIT Protocol**: Send standard frame (**Command 8**).

### 4.2.5.Motor Post-Power-Off Anti-Backdrive Protection

- **The motor’s post-power-off anti-backdrive protection can be enabled or disabled.If the motor is rotated rapidly while unpowered, a certain damping force will be felt. This protection prevents surges caused by excessively high external rotation speeds.The protection can be disabled by setting damper to 1.**

### 4.2.6. Zero Calibration Rules

- **Zero calibration is supported in CSP and Motion Control modes, but blocked in PP mode.**
- **In old firmware versions, zero calibration would cause a large deviation between the motor's target and actual position, triggering an immediate movement of the motor to the target position.**
- **In new firmware versions, when zero calibration is performed in CSP or Motion Control mode, the motor's target position is instantly updated to 0, so the motor remains stationary. Zero calibration is unavailable in PP mode.**

### 4.2.7.Position Offset

- **A position offset can be added. If an offset of 1 is set, the new zero point will be the original zero point plus 1.**
- **Some customers cannot perform zero calibration at the physical zero position due to mechanical limits.For example, if the zero point is set at a position of 1 radian, and add_offset is then set to 1, after a power cycle, the motor’s position of 1 radian will correspond to the original 1 radian position, and the motor’s zero point will be 1 radian counterclockwise from the original 1 radian position.**

### 4.2.8. The CANopen ID of the motor is consistent with the CAN ID.

- **The CANopen ID is kept the same as the CAN ID under the private protocol.**

### 4.2.9. Motor Initialization Calibration

- When the motor sets iq_test to 1, the initialization time is extended, resulting in a more accurate motor initial reference.

---

### 4.2.10. Cogging Calibration Function

- Before calibration, set iq_test to 1 and restart the motor; the motor must be under no-load conditions. Then perform cogging calibration via the host computer. After successful calibration, max_alve will be non-zero. Next, set alveolous_open to 1 to activate the function in Motion Control mode.

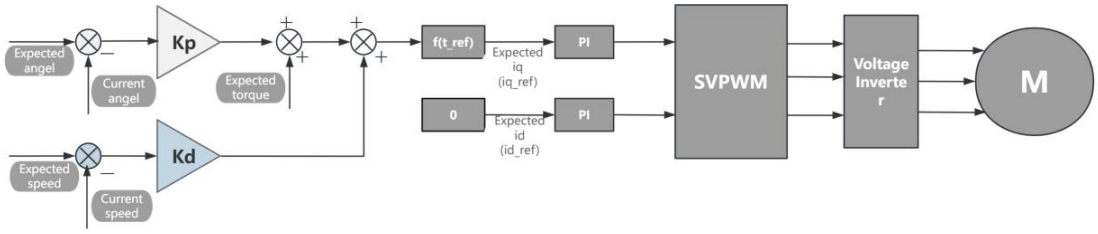
---

### 4.2.11. PP Mode Deceleration

By setting acc_stutus to 1, the acceleration and deceleration for PP mode can be configured separately. The deceleration value is stored at 0x702E. When acc_stutus is set to 0, the acceleration parameter controls both acceleration and deceleration.

## 4.3. Control mode instructions

### 4.3.1. Operation control mode

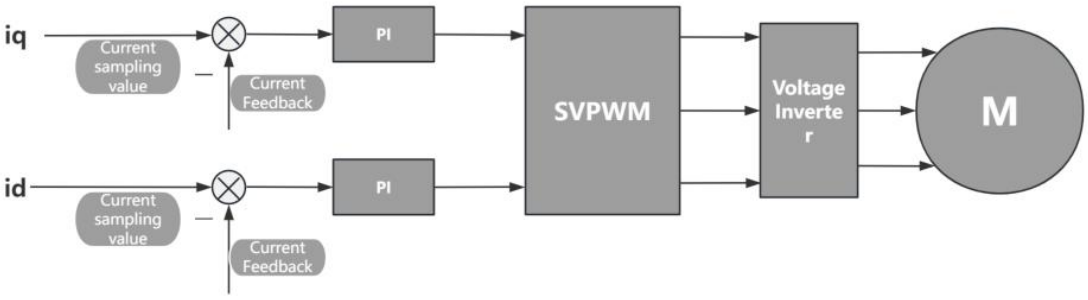


The motor is in operation control mode by default after power-on.

Send motor Enable Run frame (communication type 3) --> Send operation mode motor control command (communication type 1) --> Receive motor feedback frame (communication type 2)

Operation control mode description: The control logic of the operation and control mode is t_ref=Kd * (v_vset-v_actual)+Kp * (p_set-p_actual)+t_ff. Tref is converted to the expected iq current through an internal formula and output through the current loop Simple control demonstration: Set t_ff to 0, v_vset to 1, Kd to 1, p_set to 0, Kp to 0. If there is no external load on the motor, it will run at a speed of 1rad/s. If there is an external load, kd needs to be increased to resist the external load Set t_ff to 0, v_vset to 0, Kd to 1, p_set to 0, Kp to 0, the motor is in damping mode. When the motor is externally rotated, a damping is applied, which increases with the increase of kd. It should be noted that the motor generates electricity under this condition and requires power supply to prevent overvoltage Set t_ff to 0, v_vset to 0, Kd to 1, p_set to 5, Kp to 1. If there is no external load on the motor, it will run to the target position of 5. Increasing kp will increase the force required to maintain the target position, and kd is damping. Without kd, the motor will sway to the target position

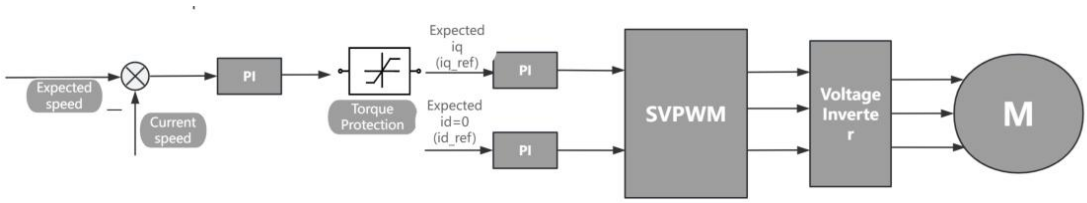
## 4.3.2. Current mode




Send motor mode parameter write command (communication type 18) Set the runmode parameter to 3 --> Send motor Enable run frame (communication type 3) --> Send motor mode parameter write command (communication

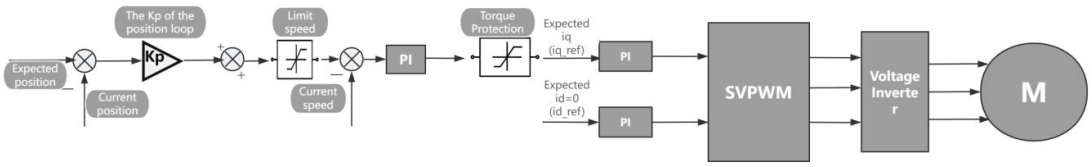
type 18) set the iq_ref parameter to the default current instruction

## 4.3.3. Velocity mode



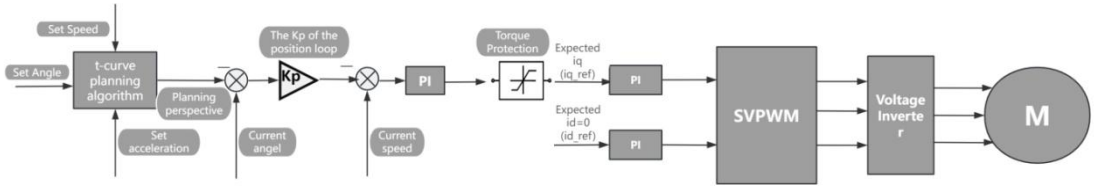

Send motor mode parameter write command (communication type 18) Set the runmode parameter to 2 --> Send motor Enable run frame (communication type 3) --> Send motor mode parameter write command (communication type 18) set limit_cur parameter as default maximum current instruction --> Send motor mode parameter write command (communication type 18) Set acc_rad parameter as default acceleration instruction --> Send motor mode parameter write command (communication type 18) Set spd_ref parameter as default speed instruction

## 4.3.4. Location Mode (CSP)




Send motor mode parameter write command (communication type 18) Set the runmode parameter to 5 --> Send motor Enable run frame (communication type 3) --> Send motor mode parameter write command (communication type 18) set limit_spd parameter as default maximum speed instruction --> Send motor mode parameter write command (communication type 18) Sets loc_ref parameter as default position instruction

## 4.3.5. Location Mode (PP)




Send motor mode parameter write command (communication type 18) Set the runmode parameter to 1 --> Send motor Enable run frame (communication type 3) --> Send motor mode parameter write command (communication type 18) set The vel_max parameter is the default maximum speed instruction --> Send motor mode parameter write command (communication type 18) Set the acc_set parameter to the default acceleration instruction --> Send motor mode parameter write command (communication type 18) Set the loc_ref parameter to the default position instruction

**Note: This mode does not support changing the speed and acceleration during operation. If you want to make an emergency stop, you can change vel_max to 0 during the process, and it will stop at the current speed and acceleration plan**

## 4.3.6. Stop running

Sending motor stop frame (communication type 4)

# 4.4. Program sample

Examples of various mode control motors are provided below (take gd32f303 as an example)

The following are library, function, and macro definitions for the various instances

```
Plain Text
```

```
#define P_MIN -12.57f

#define P_MAX 12.57f

#define V_MIN -20.0f

#define V_MAX 20.0f

#define KP_MIN 0.0f

#define KP_MAX 5000.0f

#define KD_MIN 0.0f

#define KD_MAX 100.0f

#define T_MIN -60.0f

#define T_MAX 60.0f

struct exCanIdInfo{

uint32_t id:8;

uint32_t data:16;

uint32_t mode:5;

uint32_t res:3;
```

```
};

can_receive_message_struct rxMsg;

can_trasnmit_message_struct txMsg={

.tx_sfid = 0,

.tx_efid = 0xff,

.tx_ft = CAN_FT_DATA,

.tx_ff = CAN_FF_EXTENDED,

.tx_dlen = 8,

};

#define txCanIdEx (*((struct exCanIdInfo*)&(txMsg.tx_efid)))

#define rxCanIdEx (*((struct exCanIdInfo*)&(rxMsg.rx_efid))) //将扩展帧 id 解析为自定义数据结构

int float_to_uint(float x, float x_min, float x_max, int bits){

float span = x_max - x_min;

float offset = x_min;

if(x > x_max) x=x_max;
```

```
else if(x < x_min) x= x_min;

return (int) ((x-offset)*((float)((1<<bits)-1))/span);

}

#define can_txd() can_message_transmit(CAN0, &txMsg)

#define can_rxd() can_message_receive(CAN0, CAN_FIFO1, &rxMsg)
```

The following lists the common types of communication sent:

### 4.4.1. Motor Enabled Run frame (communication type 3)

```
Plain Text
void motor_enable(uint8_t id, uint16_t master_id)
{
txCanIdEx.mode = 3;
txCanIdEx.id = id;
txCanIdEx.res = 0;
txCanIdEx.data = master_id;
txMsg.tx_dlen = 8;
txCanIdEx.data = 0;
can_txd();
}
```

### 4.4.2. Operation control mode Motor control instruction (communication type 1)

Plain Text

```
void motor_controlmode(uint8_t id, float torque, float MechPosition, float speed, float kp, float kd)

{

txCanIdEx.mode = 1;

txCanIdEx.id = id;

txCanIdEx.res = 0;

txCanIdEx.data = float_to_uint(torque,T_MIN,T_MAX,16);

txMsg.tx_dlen = 8;

txMsg.tx_data[0]=float_to_uint(MechPosition,P_MIN,P_MAX,16)>>8;

txMsg.tx_data[1]=float_to_uint(MechPosition,P_MIN,P_MAX,16);

txMsg.tx_data[2]=float_to_uint(speed,V_MIN,V_MAX,16)>>8;

txMsg.tx_data[3]=float_to_uint(speed,V_MIN,V_MAX,16);

txMsg.tx_data[4]=float_to_uint(kp,KP_MIN,KP_MAX,16)>>8;

txMsg.tx_data[5]=float_to_uint(kp,KP_MIN,KP_MAX,16);
```

```
txMsg.tx_data[6]=float_to_uint(kd,KD_MIN,KD_MAX,16)>>8;

txMsg.tx_data[7]=float_to_uint(kd,KD_MIN,KD_MAX,16);

can_txd();

}
```

### 4.4.3. Motor mode parameter write command (communication type 18, running mode switch)

Plain Text

```
void motor_reset(uint8_t id, uint16_t master_id)

{

txCanIdEx.mode = 4;

txCanIdEx.id = id;

txCanIdEx.res = 0;

txCanIdEx.data = master_id;

txMsg.tx_dlen = 8;

for(uint8_t i=0;i<8;i++)

{
```

```
txMsg.tx_data[i]=0;

}

can_txd();

}
```

### 4.4.4. Motor mode parameter write command (communication type 18, running mode switch)

```
Plain Text
uint8_t runmode;

uint16_t index;

void motor_modechange(uint8_t id, uint16_t master_id)

{

txCanIdEx.mode = 0x12;

txCanIdEx.id = id;

txCanIdEx.res = 0;

txCanIdEx.data = master_id;

txMsg.tx_dlen = 8;
```

```
for(uint8_t i=0;i<8;i++)

{

txMsg.tx_data[i]=0;

}

memcpy(&txMsg.tx_data[0],&index,2);

memcpy(&txMsg.tx_data[4],&runmode, 1);

can_txd();

}
```

### 4.4.5. Motor mode parameter write command (communication type 18, control parameter write)

Plain Text

```
uint16_t index;

float ref;

void motor_write(uint8_t id, uint16_t master_id)

{
```

```
txCanIdEx.mode = 0x12;

txCanIdEx.id = id;

txCanIdEx.res = 0;

txCanIdEx.data = master_id;

txMsg.tx_dlen = 8;

for(uint8_t i=0;i<8;i++)

{

txMsg.tx_data[i]=0;

}

memcpy(&txMsg.tx_data[0],&index,2);

memcpy(&txMsg.tx_data[4],&ref,4);

can_txd();

}
```

# 5. Explanation of Canopen Communication Protocol Types

## 5.1. Introduction to CANopen Communication

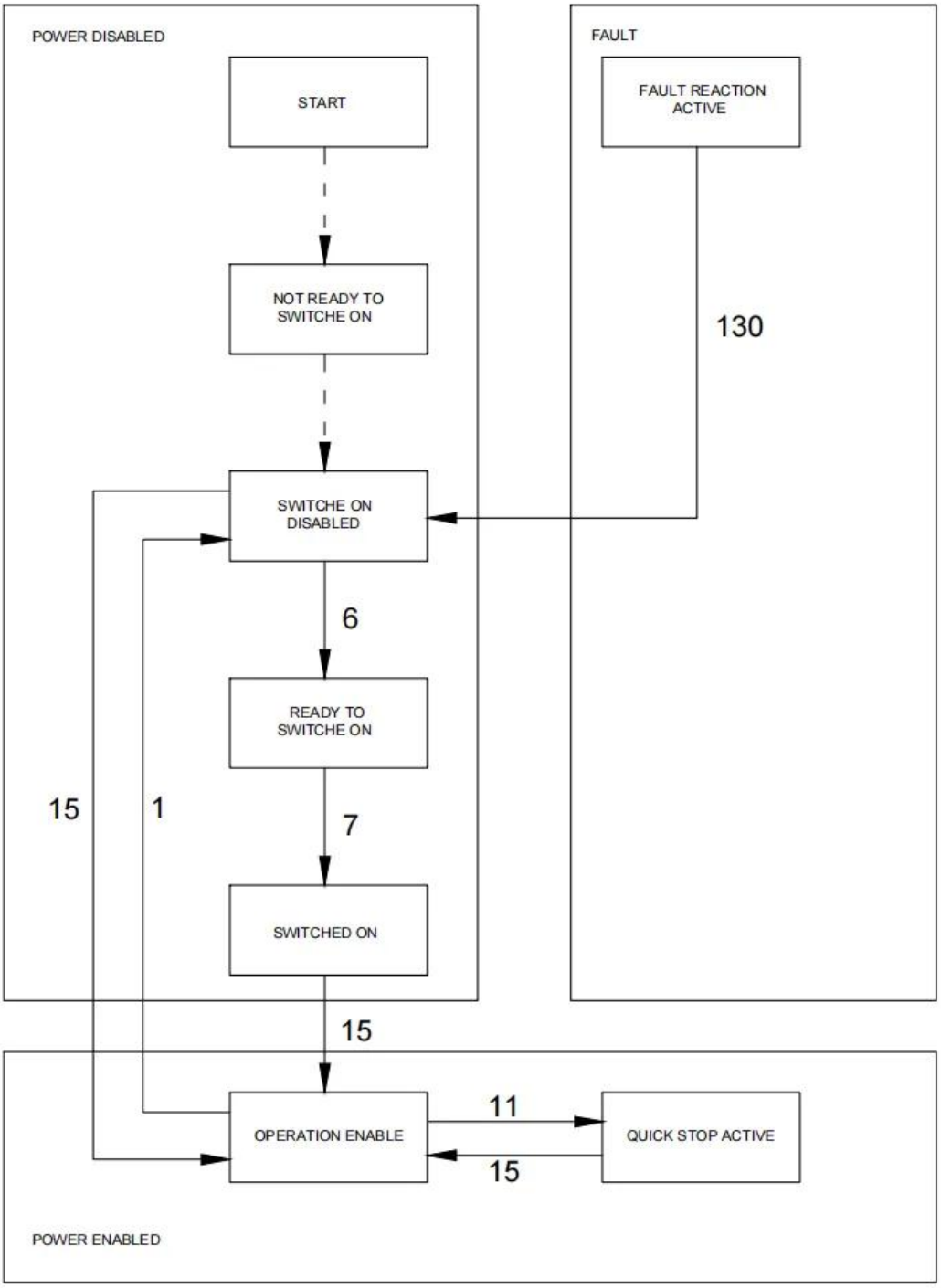
CANopen is a "higher-level protocol" based on the CAN bus. This means that the CAN bus (ISO 11898) acts like a container truck, serving as the "transportation vehicle" for CANopen information. CAN simply implements the transmission of frames with an 11-bit CAN ID, a Remote Transmission (RTR) bit, and 64 data bits (related to higher-level protocols). The CAN bus plays the same role in CANopen as it does in the J1939 protocol. CANopen implements Layer 7 of the OSI model and is compatible with other data link layer protocols besides CAN.Please download the necessary EDS files for CANopen from the official website download center at www.robstride.com.

## 5.2. Canopen Protocol Message Classification

| Classification | Function |
|---|---|
| NMT Network Management Messages | Manage the network and switch node states. NMT network management messages are typically sent by the master station. |
| SDO Service Data Object Message | Used for setting device parameters or transmitting critical data. Typically, the master station initiates an SDO message, and the slave station responds. However, a slave station can also initiate an SDO message, with the master station responding, for example, in the transmission of critical data. |
| PDO (Process Data Object) messages | Transmit process data from various devices, such as temperature and speed. Both the master and slave stations send these messages. |
| EMCY Emergency Message | Transmits fault information about the transmission equipment. Both the master and slave stations send this message. |
| SYNC Synchronization Message | Synchronization data, used to synchronize TPDO data on the slave station. Generally sent by the master station. |

| Classification | Function |
| --- | --- |
| NODE GUARDING Message | The master station requests the slave station's status; the master station queries, and the slave station responds. |
| Heartbeat Message | A device actively sends a heartbeat to indicate that it is online. Both the master and slave devices can send heartbeats. |

## 5.3. State Machine Description




**Motor Enable**:

When initially powered on, the motor defaults to the **SWITCH_ON_DISABLED** state. To transition to **OPERATION_ENABLE**, modify the **Controlword (6040H)** to **6, 7, or 15** (step-by-step transition), or directly set it to **15** for immediate

enablement.

**Stopping the Motor**:

If the motor is in **OPERATION_ENABLE** state and needs to stop normally, modify the **Controlword (6040H)** to **1**. The motor will return to the disabled state (**SWITCH_ON_DISABLED**).

**Emergency Stop (Use with Caution—Risk of Voltage Surge)**:

During operation, an emergency stop can be triggered by setting the **Controlword (6040H)** to **11**.

**Fault Clearance**:

If the motor enters a **FAULT** state due to protection mechanisms, modifying the **Controlword (6040H)** can clear standard errors.

**Motor Watchdog**

Set the CAN communication watchdog via index 1 of object 0x6099. After configuration, a value of 20000 represents 1 second. If no communication frame is received within the set time, the motor will be disabled. The default value is 0 (disabled), and a non-zero value enables the watchdog.

**Important Note**:

Mode changes for this motor must be performed in the **disabled state (SWITCH_ON_DISABLED)**. Ensure the desired mode is configured **before** enabling **OPERATION_ENABLE** to avoid unexpected behavior.

## 5.4. Status Feedback Parameters

| Index | Name | Attribute | Type | Unit |
|---|---|---|---|---|
| **603F** | **Error_code** | Read-only | UINTEGER16 | / |
| **6041** | **Statusword** | Read-only | UINTEGER16 | / |
| **6061** | **Modes_of_operation_display** | Read-only | INTEGER8 | / |
| **6062** | **Position_demand_value** | Read-only | INTEGER32 | Pulses (1 rev = 16,384 pulses) |

| Index | Name | Attribute | Type | Unit |
|---|---|---|---|---|
| **6064** | **Position_actual_value** | Read-only | INTEGER32 | Pulses (1 rev = 16,384 pulses) |
| **606B** | **Velocity_demand_value** | Read-only | INTEGER32 | 0.1 rpm |
| **606C** | **Velocity_actual_value** | Read-only | INTEGER32 | 0.1 rpm |
| **6077** | **Torque_actual_value** | Read-only | INTEGER16 | 0.1% load ratio (1000 = 20 N·m) |
| **6078** | **Current_actual_value** | Read-only | INTEGER16 | mA |
| **6079** | **DC_link_circuit_voltage** | Read-only | INTEGER32 | mV |

## 5.5. Homing Mode (Zero Position Setting)

| Index | Name | Attribute | Type | Unit |
|---|---|---|---|---|
| **6040** | **Controlword** | Read-write | UINTEGER16 | / |
| **6060** | **Modes of operation** | Read-write | INTEGER8 | / |

**Homing method**:

- Set **Modes of operation** to **6** while the motor is in the **disabled state (SWITCH_ON_DISABLED)**. The motor will then define the current position as the zero point.
- To **hold the zero position**, modify the **Controlword** to **15**, and the motor will maintain its position at the home location.

## 5.6. Position Mode (PP - Profile Position)

| Index | Name | Attribute | Type | Unit |
|---|---|---|---|---|
| **6040** | **Controlword** | Read-write | UINTEGER16 | / |
| **6060** | **Modes of operation** | Read-write | INTEGER8 | / |
| **6067** | **Position_window** | Read-write | UINTEGER32 | Pulses (1 rev = 16,384 pulses) |
| **6068** | **Position_window_time** | Read-write | UINTEGER16 | ms |
| **6071** | **Target_torque** | Read- | INTEGER16 | 0.1% load ratio (1000 = 20 |

| Index | Name | Attribute | Type | Unit |
|---|---|---|---|---|
| | | write | | N·m) |
| **607A** | **Target_position** | Read-write | INTEGER32 | Pulses (1 rev = 16,384 pulses) |
| **6081** | **Profile_velocity** | Read-write | UINTEGER32 | 0.1 rpm |
| **6083** | **Profile_acceleration** | Read-write | UINTEGER32 | 0.1 rpm/s |
| | | | | |

**Steps to Configure Position Mode (PP):**

1. While the motor is in the **disabled state (SWITCH_ON_DISABLED)**, set **Modes of operation** to **1**.
   - **Mandatory parameters**:
     - **Target_torque** (absolute max torque in position mode)
     - **Profile_velocity** (absolute speed in position mode)
     - **Profile_acceleration** (absolute acceleration in position mode)
   - **Optional parameters**:
     - **Position_window** (if not set, window check is disabled)
     - **Position_window_time** (if not set, window check is disabled)
2. Set **Controlword (6040)** to **15** to enable operation.
3. **Set Target_position** (absolute position) to move the motor to the desired position.

## 5.7. Position Mode (CSP - Cyclic Synchronous Position)

| Index | Name | Attribute | Type | Unit |
|---|---|---|---|---|
| **6040** | **Controlword** | Read-write | UINTEGER16 | / |
| **6060** | **Modes of operation** | Read- | INTEGER8 | / |

| Index | Name | Attribute | Type | Unit |
|---|---|---|---|---|
| | | write | | |
| **6067** | **Position_window** | Read-write | UINTEGER32 | Pulses (1 rev = 16,384 pulses) |
| **6068** | **Position_window_time** | Read-write | UINTEGER16 | ms |
| **6071** | **Target_torque** | Read-write | INTEGER16 | 0.1% load ratio (1000 = 20 N·m) |
| **607A** | **Target_position** | Read-write | INTEGER32 | Pulses (1 rev = 16,384 pulses) |
| **6081** | **Profile_velocity** | Read-write | UINTEGER32 | 0.1 rpm |

**Steps to Configure Position Mode (CSP):**

1. While the motor is in the **disabled state (SWITCH_ON_DISABLED)**, set **Modes of operation** to **5**.
   - **Mandatory parameters**:
     - **Target_torque** (absolute max torque in position mode)
     - **Profile_velocity** (absolute speed in position mode)
   - **Optional parameters**:
     - **Position_window** (0 = disabled)
     - **Position_window_time** (0 = disabled)
2. Set **Controlword (6040)** to **15** to enable operation.
3. **Set Target_position** (absolute position) to move the motor to the desired position.

## 5.8. Velocity Mode

| Index | Name | Attribute | Type | Unit |
|---|---|---|---|---|
| **6040** | **Controlword** | Read-write | UINTEGER16 | / |
| **6060** | **Modes of operation** | Read-write | INTEGER8 | / |
| **6071** | **Target_torque** | Read-write | INTEGER16 | 0.1% load ratio (1000 = 20 |

| Index | Name | Attribute | Type | Unit |
|---|---|---|---|---|
| | | | | N·m) |
| **60FF** | **Target_velocity** | Read-write | INTEGER32 | 0.1 rpm |

**Steps to Configure Velocity Mode:**

1. While the motor is in the **disabled state (SWITCH_ON_DISABLED)**, set **Modes of operation** to **3**.
   - **Mandatory parameter**:
     - **Target_torque** (absolute max torque in velocity mode)
2. Set **Controlword (6040)** to **15** to enable operation.
3. **Set Target_velocity** to reach the desired speed.

## 5.9. Torque Mode

| Index | Name | Attribute | Type | Unit |
|---|---|---|---|---|
| **6040** | **Controlword** | Read-write | UINTEGER16 | / |
| **6060** | **Modes of operation** | Read-write | INTEGER8 | / |
| **6071** | **Target_torque** | Read-write | INTEGER16 | 0.1% load ratio (1000 = 20N·m) |

**Steps to Configure Torque Mode:**

1. While the motor is in the **disabled state (SWITCH_ON_DISABLED)**, set **Modes of operation** to **4**.
2. Set **Controlword (6040)** to **15** to enable operation.
3. **Set Target_torque** to output the desired torque.

## 5.10. Protocol Switching (Extended Frame): Switch Motor Protocol (Takes Effect After Power Cycle)

| Data Field | 29-bit ID | 8-Byte Data Area |
|---|---|---|

| | | |
|---|---|---|
| **Size** | Bit 28~0 | Byte 0~6 |
| **Description** | **0xFFF** | 01 02 03 04 05 06 F_CMD |

- **F_CMD** (Byte 6) defines the motor protocol:
  - **0**: Private protocol (default)
  - **1**: CANopen protocol
  - **2**: MIT protocol

**Response Frame:**

| Data Field | 11-bit ID | 8-Byte Data Area |
|---|---|---|
| **Size** | Bit 10~0 | Byte 0~7 |
| **Description** | Motor ID | 64-bit MCU unique identifier |

# 6. MIT Communication Protocol Description

The motor communication adopts the CAN 2.0 interface with a default baud rate of 1 Mbps. The baud rate can be modified by switching to the private protocol. The standard frame format is as follows:

| Data Field | 11-bit ID | | 8-byte Data Area |
|---|---|---|---|
| **Size** | Bit 10~8 | Bit 7~0 | Byte 0~7 |
| **Description** | Mode type | ID | |

- Supported Control Modes:
- **MIT Mode**: Provides five motion control parameters to the motor.
- **Velocity Mode**: Specifies the target speed for the motor.
- **Position Mode**: Specifies the target position and speed, allowing the motor to run to the designated position at the configured speed.

## 6.1. Response Command 1: Data Feedback (Motor Status)

| Data Field | 11-bit ID | 8-byte Data Area |
|---|---|---|
| **Size** | Bit 10~0 | Byte 0~7 |
| **Description** | Host ID | Byte 0: Motor CAN ID<br>Byte 1~2: The current angle [0~65535], corresponds to (-12.57 rad ~ 12.57 rad)<br>Byte 3 (high 8 bits),Byte 4[7-4] (low 4 bits): The current speed [0~4096], corresponds to (-33 rad/s ~ 33 rad/s)<br>Byte 4[3-0] (high 4 bits),Byte 5 (low 8 bits): The current torque [0~4096], corresponds to (-60 N·m ~ 60 N·m)<br>Byte 6~7: Winding temperature (in degrees): Temp(Celsius) *10 |

## 6.2. Response Command 2: MCU Identification

| Data Field | 11-bit ID | 8-byte Data Area |
|---|---|---|
| **Size** | Bit 10~0 | Byte 0~7 |

| | | |
|---|---|---|
| **Description** | Motor ID | 64-bit MCU unique identifier |

## 6.3. Command 1: Enable Motor Operation

| Data Field | 11-bit ID | 8-byte Data Area |
|---|---|---|
| **Size** | Bit 10~0 | Byte 0~7 |
| **Description** | Target motor CAN ID | FF FF FF FF FF FF FF FC |

**Response:** Response Command 1

## 6.4. Command 2: Stop Motor Operation

| Data Field | 11-bit ID | 8-byte Data Area |
|---|---|---|
| **Size** | Bit 10~0 | Byte 0~7 |
| **Description** | Target motor CAN ID | FF FF FF FF FF FF FF FD |

**Response:** Response Command 1

## 6.5. Command 3: MIT Dynamic Parameters

| Data Field | 11-bit ID | 8-byte Data Area |
|---|---|---|
| **Size** | Bit 10~0 | Byte 0~1: Target angle [0~65535], (-12.57 rad ~ 12.57 rad)Byte 2 (high 8 bits), Byte 3[7-4] (low 4 bits): Target speed [0~4096], (-33 rad/s ~ 33 rad/s)Byte 3[3-0] (high 4 bits), Byte 4 (low 8 bits): Kp [0~4096], (0~500)Byte 5 (high 8 bits), Byte 6[7-4] (low 4 bits): Kd [0~4096], (0~5)Byte 6[3-0] (high 4 bits), Byte 7 (low 8 bits): Target torque [0~4096], (-60 N·m ~ 60 N·m) |

**Response:** Response Command 1

## 6.6. Command 4: Set Zero Position (Non-Position Mode)

| Data Field | 11-bit ID | 8-byte Data Area |
|---|---|---|
| **Size** | Bit 10~0 | Byte 0~7 |
| **Description** | Target motor CAN ID | FF FF FF FF FF FF FF FE |

**Response:** Response Command 1

## 6.7. Command 5: Clear Errors & Read Fault Status

| Data Field | 11-bit ID | 8-byte Data Area |
|---|---|---|
| **Size** | Bit 10~0 | FF FF FF FF FF FF F_CMD FBF_CMD:- 0xFF → Clear current fault- Any other value → Returns fault value in Byte 1 of the response |

**Response (Fault Clear):** Response Command 1

**Fault Status Response:**

| Data Field | 11-bit ID | 8-byte Data Area |
|---|---|---|
| **Size** | Bit 10~0 | Byte 0: Motor CAN IDByte 1~4: Fault value (Non-zero: Fault present; 0: Normal)Bit 14: Stall/I²t overload fault<Bit 7: Encoder not calibrated<br>Bit 3: Overvoltage faultBit 2: Undervoltage faultBit 1: Driver IC faultBit 0: Motor overtemperature fault (Default threshold: 145°C) |

## 6.8. Command 6: Set Operation Mode

| Data Field | 11-bit ID | 8-byte Data Area |
|---|---|---|
| **Size** | Bit 10~0 | FF FF FF FF FF FF F_CMD FCF_CMD: Mode type- 0: MIT mode (default)- 1: Position mode- 2: Velocity mode |

**Response:** Response Command 1

## 6.9. Command 7: Modify Motor CAN ID

| Data Field | 11-bit ID | 8-byte Data Area |
|---|---|---|
| **Size** | Bit 10~0 | FF FF FF FF FF FF F_CMD FAF_CMD: Target motor CAN ID |

**Response:** Response Command 2

## 6.10. Command 8: Change Communication Protocol (Takes Effect After Power Cycle)

| Data Field | 11-bit ID | 8-byte Data Area |
|---|---|---|
| **Size** | Bit 10~0 | FF FF FF FF FF FF F_CMD FDF_CMD: Protocol type- 0: Private protocol (default)- 1: CANopen<br>- 2: MIT protocol |

**Response:** Response Command 2

## 6.11. Command 9: Modify Host CAN ID

| Data Field | 11-bit ID | 8-byte Data Area |
|---|---|---|
| **Size** | Bit 10~0 | FF FF FF FF FF FF F_CMD 01F_CMD: Host CAN ID |

**Response:** Response Command 2

## 6.12. Command 10: Position Mode Control Command

| Data Field | 11-bit ID | | 8-byte Data Area |
|---|---|---|---|
| **Size** | Bit 10~8 | Bit 7~0 | Byte 0~3: Target position (rad, 32-bit float)Byte 4~7: Target speed (rad/s, 32-bit float) |
| **Description** | 1 | Target motor CAN ID | |

**Response:** Response Command 1

## 6.13. Command 11: Velocity Mode Control Command

| Data Field | 11-bit ID | | 8-byte Data Area |
|---|---|---|---|
| **Size** | Bit 10~8 | Bit 7~0 | Byte 0~3: Target speed (rad/s, 32-bit float) Byte 4~7: Current limit in speed/position mode (A, 32-bit float) |
| **Description** | 2 | Target | |

| | | | |
|---|---|---|---|
| | | motor CAN ID | |

**Response:** Response Command 1

## 6.14. Command 12:Save motor data (requires firmware update to 0.3.1.41)

| Data Field | 11-bit ID | 8-byte Data Area |
|---|---|---|
| **Size** | Bit 10~0 | Byte0~Byte7 |
| **Description** | Target motor CANID | FF FF FF FF FF FF FF F8 |

Response frame: Response command 2

## 6.15. Command 13: Motor actively reports (requires firmware update to 0.3.1.41)

| Data Field | 11-bit ID | 8-byte Data Area |
|---|---|---|
| **Size** | Bit 10~0 | Byte0~Byte7 |
| **Description** | Target motor CANID | FF FF FF FF FF FF F_CMD F9<br>The F_CMD byte indicates the motor protocol type.<br>0 indicates no reporting (default).<br>1 is for reporting |

Response frame: Response command 1

## 6.16. Command 14: Read motor parameters (requires firmware update to 0.3.1.41)

| Data Field | 11-bit ID | | 8Byte 数据区 |
|---|---|---|---|
| **Size** | bit10~8 | bit7~0 | Byte0~Byte7 |
| | 3 | 7F | 05 70 00 00 00 00 00 00 |
| **Description** | 3 | Target motor CANID | Byte0~1: index(See the read/write parameter table below for details.)<br>Byte2~3: 00<br>Byte4~7: 00 The above data is in low byte first, high byte last. |

Response frame:

| Data Field | 11-bit ID | | 8Byte 数据区 |
|---|---|---|---|
| **Size** | bit10~8 | bit7~0 | Byte0~Byte7 |
| | 3 | 7F | 05 70 00 00 00 00 00 00 |
| **Description** | 3 | Target motor CANID | Byte0~1: index See the read/write parameter table for details (same as the private protocol parameter table).<br>Byte2~3: 00<br>Byte4~7: 00 For parameter data, data of 1 byte or more is formatted with the least significant byte first and the most |

| | | | |
|---|---|---|---|
| | | | significant byte last. |

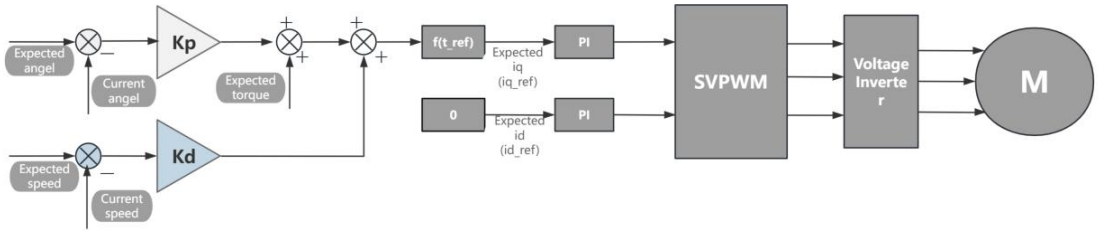
## 6.17. Command 15: Write motor parameters (requires firmware update to 0.3.1.41)

| Data Field | 11-bit ID | | 8Byte 数据区 |
|---|---|---|---|
| **Size** | bit10~8 | bit7~0 | Byte0~Byte7 |
| | 4 | 7F | 29 70 00 00 00 00 00 00 |
| **Description** | 4 | Target motor CANID | Byte0~1: index See the read/write parameter table for details (same as the private protocol parameter table).<br>Byte2~3: 00<br>Byte4~7: 00 Parameter data: The above data is in low byte first, high byte last. |

Response frame:

| Data Field | 11-bit ID | | 8Byte 数据区 |
|---|---|---|---|
| **Size** | bit10~8 | bit7~0 | Byte0~Byte7 |
| | 4 | 7F | 29 70 00 00 00 00 00 00 |
| **Description** | 4 | Target motor CANID | Byte0~1: index See the read/write parameter table for details (same as the private protocol parameter table).<br>Byte2~3: 00<br>Byte4~7: 00 Parameter data: The above data is in low byte first, high byte last. |

## 6.18. Motion Control Mode



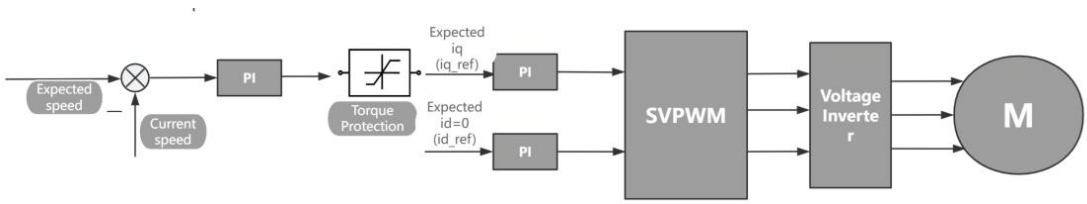

The motor defaults to Motion Control Mode upon power-up.

1. Send the *Motor Enable Command* (Command 1).

2. Send the *Motion Control Command* (Command 3) to activate dynamic parameter control.

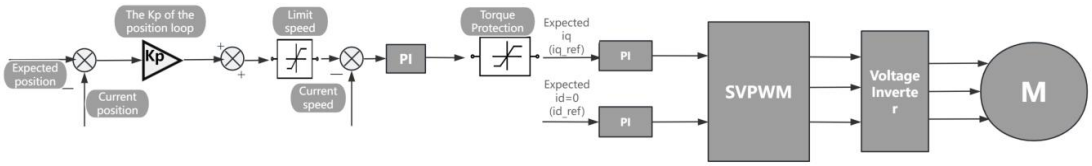
3. Send the *Motor Stop Command* (Command 2) to halt operation when needed.

## 6.19. Velocity Mode




1. Configure the motor's operation mode by sending *Set Operation Mode Command* (Command 6) with **Mode = 2 (Velocity Mode)**.
2. Send the *Motor Enable Command* (Command 1) to activate the motor.
3. Send the *Velocity Mode Control Command* (Command 11) to set the **maximum current (absolute value)** and **target speed**.
4. To stop, send the *Motor Stop Command* (Command 2).

## 6.20. Position Mode (CSP - Cyclic Synchronous Position)




1. Configure the motor's operation mode by sending *Set Operation Mode Command* (Command 6) with **Mode = 1 (Position Mode)**.
2. Send the *Motor Enable Command* (Command 1) to activate the motor.
3. Send the *Position Mode Control Command* (Command 10) to set the **maximum speed (absolute value)** and **target position**.
4. To stop, send the *Motor Stop Command* (Command 2).

# 7. Version History

| Version Number | Description | Date |
|---|---|---|
| 1.0 | Initial Release | November 25, 2025 |